Journal of Accounting Research

DOI: 10.1111/1475-679X.70011
Journal of Accounting Research
Vol. 64 No. 1 March 2026
Printed in U.S.A.

CHICAGO BOOTH

Amendment Thresholds and Voting Rules in Debt Contracts

JUDSON CASKEY,* KANYUAN HUANG,†
AND DANIEL SAAVEDRA‡

Received 21 November 2022; accepted 29 April 2025

ABSTRACT

Most loan contracts in the United States contain a provision for lender voting rules. We study the optimal voting rule that allows lenders to waive a covenant violation. When lenders have heterogeneous preferences, lenient voting rules increase the probability of waivers that allow inefficient investments. Stringent voting rules tend to allocate the marginal vote to lenders who deny waivers after false alarms so that they can renegotiate the loan to extract value from the firm, which incurs deadweight costs. In equilibrium, the optimal voting rule balances these two forces to improve contracting efficiency. We derive and empirically test comparative statics on how the optimal voting rule varies with lenders' preferences and the borrower's accounting properties. Our model

*Mitch Daniels School of Business, Purdue University; †The Chinese University of Hong Kong, Shenzhen; ‡Unaffiliated

Accepted by Valeri Nikolaev. We appreciate comments from the associate editor, the anonymous referee, Ying Cao, Richard Carrizosa, Jacky Chau, John Donovan (Discussant), Henry Friedman, Mark Grinblatt, Zhaoyang Gu, Mingyi Hung, Shohini Kundu, Ningzhong Li (Discussant), Xinlei Li, Francis Longstaff, Rahul Menon, Arthur Morris, Han Stice, Avanidhar Subrahmanyam, Ivo Welch, Chunmei Zhu, and participants at Accounting and Economics Society's Winter Retreat, American Accounting Association Annual Conference, CUHK, CUHK (Shenzhen), Egyptian Online Seminars in Business, Accounting, and Economics, Hawaii Accounting Research Conference, HKUST, University of Iowa, The Ohio State University, Purdue University, UCLA, UC Riverside, University of Iowa, and University of Leeds. The authors have no conflicts of interest to declare. An online appendix to this paper can be downloaded at https://www.chicagobooth.edu/jar-online-supplements.

offers a rationale for the prevalent use of voting rule clauses in syndicated loan contracts.

JEL codes: G21, G28, G32, H25, H32

Keywords: amendment thresholds; voting rules; covenant waivers; agency problems; bank loans; renegotiation

1. Introduction

Most syndicated loan agreements in the United States specify required lenders clauses that determine lender voting rules to waive a covenant violation.[1] The following is a typical voting clause:

> No amendment or waiver of any provision of this Agreement or any other Loan Document, and no consent with respect to any departure by the Company therefrom, shall be effective unless the same shall be in writing and signed by the Majority Banks [later defined as holders of at least 51% of the outstanding principal].[2]

In the event of a covenant violation, the required lenders behave as gatekeepers between the firm and the entire group of lenders—they decide whether to waive the covenant or subject the firm to a renegotiation that lenders can use to extract concessions such as accelerated payments, higher interest rates, or reduced loan commitments. Roberts and Sufi [2009a] document that 63% of the covenant violations are waived by creditors without altering major loan terms.[3]

While a large body of literature examines the frequency and economic consequences of covenant violations (e.g., Beneish and Press [1993], Chen and Wei [1993], Smith [1993], DeFond and Jiambalvo [1994], Sweeney [1994], Chava and Roberts [2008], Nini, Smith, and Sufi [2012], Vashishtha [2014]), we are not aware of prior research examining the voting rules to waive covenant violations. This study fills this gap in the literature

[1] These clauses are sometimes referred to as "required banks," "requisite lenders," "requisite banks," "majority lenders," or "majority banks" in the loan contract. Hereafter, we use the term voting rules.

[2] The excerpt is from Section 10.1 of a contract between Chase Bank, the lead lender, and General Mills. As we discuss further in section 2, terms such as the interest rate, payment schedule, and loan commitment are sacred rights that may change only with unanimous consent by all lenders. Required lenders clauses apply to terms such as a covenant waivers. The full contract can be accessed at the following URL: https://www.sec.gov/Archives/edgar/data/40704/000089710101500115/generalmills010796_ex99-1.txt

[3] Roberts and Sufi [2009a, p. 1688] examine SEC filings of a random sample of 500 covenant violators. In 63% of the cases, the company reports a waiver and discusses no further loan modifications. As Roberts and Sufi note, firms do not have to disclose the exact nature of the waiver. It is therefore possible that nondisclosure stems from the waiver entailing no material concessions, or that the waiver contains material concessions that the firm failed to disclose in violation of the SEC's materiality rule (Caskey, Huang, and Saavedra [2023], Li, Neamtiu, and Tu [2024]). Waivers in our model refer to a lack of material concessions.

by examining the following: (1) Why do most syndicated loan contracts use a required lenders clause? (2) How are voting rules in those clauses determined?

We build a model of the optimal voting rule in the presence of lenders with heterogeneous preferences. Specifically, we extend Gârleanu and Zwiebel's [2009] model by adding a contractable signal that can trigger covenant violations, and a group of lenders with heterogeneous preferences to waive covenant violations. The contractable signal provides noisy information about a binary state of whether investment would be profitable (Good) or unprofitable (Bad). All parties can observe, but cannot contract on, the true state. In the event of a covenant violation, the voting rule specifies the fraction of lenders that can waive the covenant to allow investment to proceed without renegotiation. If these lenders do not waive the covenant, the firm can invest only if it renegotiates the loan, which requires agreement by all lenders and thereby allows lenders to extract the value of any positive net present value (NPV) investment by, for example, increasing the loan spread or modifying other loan terms.

We model lenders with heterogeneous preferences for whether to allow the firm to invest in new projects without extracting concessions (hereafter "private benefits").[4] For example, some lenders might waive covenant violations to maintain good relations with the borrower and thereby improve their chances of future business (Drucker and Puri [2005], Bharath et al. [2007]). Contracting parties face uncertainty regarding lenders' private benefits, which leads to uncertainty about potential waiver outcomes. A covenant violation can reflect a true positive indicating that investment would be unprofitable (Bad state), or can be a false alarm or Type I error described by Gigler et al. [2009] as covenant violations when investment would be profitable (Good state). For a given voting threshold, there is a chance that the marginal voter prefers to deny waivers and extract surplus from the borrower when the covenant is tripped in the Good state, which we refer to as the lender having Low private benefits. Conversely, there is a chance that the marginal voter prefers to waive covenant violations and allow inefficient investments in the Bad state, which we refer to as the lender having High private benefits.

Because lenders' willingness to waive covenants increases with their private benefits, it creates the following tradeoff: A strict voting rule incurs efficiency costs due to false alarms in Good states, while an overly lenient threshold waives a covenant violation in Bad states, leading to excessive investments. The second potential efficiency loss—waivers in the Bad state—differs from the Type II errors in which a Bad state fails to trigger a covenant violation.[5] Instead, it arises from the marginal lender's private incentive to

[4] The voting rule is arbitrary if lenders have homogeneous preferences because each individual lender will choose the same course of action as the entire group.

[5] Several prior studies examine Type II errors (e.g., Watts [2003], Gigler et al. [2009], Demerjian, Owens, and Sokolowski [2023], Zhu [2024]). Our model also incorporates this

maintain a good relationship with the borrower, at the expense of other syndicate members.

The optimal voting rule maximizes contracting efficiency by balancing the expected costs of false alarms against the expected costs of inefficient waivers. If there were no uncertainty about lenders' private benefits, a voting rule could fully eliminate inefficiencies by ensuring the marginal vote resides with a lender having a "Goldilocks"-type level of private benefits—high enough to waive covenants for profitable investments but low enough to enforce covenants to block unprofitable investments. When there is uncertainty about lenders' private benefits, the optimal voting rule will typically be more stringent than what would maximize the likelihood of a Goldilocks lender having the marginal vote. This is because, under reasonable assumptions about payoffs and the quality of the contractable signal, inefficiencies from excessive investments in Bad states outweigh the costs of renegotiation in Good states. We also show that these conditions guarantee that a contract with covenant-based contingent control yields lower efficiency losses than one without covenants.

Our model's comparative statics predict that voting thresholds vary with the level of lenders' private benefits. Ceteris paribus, voting thresholds increase in lenders' private benefits. Lenders with high private benefits waive covenants in states where investment is inefficient, which reduces shareholders' ex ante payoffs. A higher voting threshold reduces this cost by making it less likely that the marginal voter has high private benefits. When many syndicate members have low private benefits, lenient voting rules reduce the likelihood that lenders extract concessions from borrowers after a noisy covenant metric triggers a false alarm. Additionally, we show that, under certain distributional assumptions, the optimal voting threshold increases with uncertainty about lenders' private benefits, and shareholders' expected payoffs decrease with uncertainty. This is because uncertainty increases the expected costs from excessive investments more than it increases expected renegotiation costs.

We also analyze how voting rules interact with accounting conservatism. Gigler et al. [2009] predict that conservative accounting detracts from the efficiency of lending agreements because, when a project's ex ante value exceeds its liquidation value, the cost of false alarms outweighs the cost of "undue optimism" that allows unprofitable investment. The optimal voting rule tends to be more lenient when accounting is conservative, allowing the syndicate to waive a covenant violation without concessions from low private benefit lenders. This mitigates the cost of false alarms so that conservative accounting increases contract efficiency.

---

feature but it does not affect tradeoff in deciding the optimal voting rule. The voting rule is irrelevant following Type II errors because these errors lead to shareholders offering renegotiated loan terms. These changes directly affect the lenders' payment terms and therefore require unanimous lender approval. We further discuss this in section 3.

Lastly, we examine how the optimal voting rule interacts with the number of performance-based signals, which resembles performance covenants in loan contracts (Demerjian [2011], Christensen and Nikolaev [2012]). When a contract uses multiple covenant signals, the optimal voting rule tends to be more lenient to offset the increased probability of false alarms. The intuition is similar to the prediction for accounting conservatism—contracts with more covenants have a greater likelihood of triggering a false alarm and a lenient voting rule reduces the expected costs of false alarms.

We empirically test our predictions using a sample of over 17,000 syndicated loans in Dealscan and two measures of the contract amendment threshold that, respectively, correspond to the fraction of *dollars* and to the minimum fraction of *lenders* required to amend a contract. The measures differ because lenders' loan commitments differ. The dollar-based first measure is specified in the loan contract's required lenders clause. Approximately 75% of syndicated loan contracts in the United States set this threshold to 51%, that is, the contract requires the group of lenders holding at least 51% of the loan to approve a covenant waiver. The remaining contracts set thresholds above 51%, primarily at 66.7%. Lenders' loan commitments and the dollar-based threshold determine the lender-based second measure.

Our tests provide evidence that voting rules vary with lenders' private benefits. Specifically, we find voting rules are more stringent when there are prior underwriting relationships between the borrower and syndicate lenders, which proxies for syndicates including lenders that more strongly favor waivers (Drucker and Puri [2005], Fernando et al. [2012]). We also find more lenient thresholds for syndicates that include hedge funds, where we take hedge funds' tendency to be hard bargainers in renegotiations as an indicator that they prefer renegotiating for better terms over waiving covenants. We find support for our prediction that contracts have more lenient voting rules when they have more performance covenants. This is consistent with using relatively lenient voting rules to mitigate efficiency losses from false alarms associated with multiple covenants. However, we do not find a statistically significant association between accounting conservatism and voting rules.[6]

Our paper contributes to three areas of research. First, it contributes to the literature on covenant violations and loan renegotiations by developing a model of the optimal voting rule. Prior literature examines the determinants of loan renegotiations (Roberts [2015], Nikolaev [2018], and Dou [2020]), as well as their consequences (Asquith et al. [2005], Denis and Wang [2014], Li et al. [2016], Demerjian [2017], and Saavedra [2018]). Bolton and Scharfstein [1996] analytically show that lenders can use voting

[6] This may be due to conservative accounting affecting behaviors that are not in our model but do impact the voting rule (see Watts [2003] for examples). For instance, conservatism can increase earnings manipulation (Bertomeu, Darrough, and Xue [2017]), which could lead to a more stringent voting rule.

rules to reduce the costs of strategic default by borrowers.[7] Luneva [2025] uses a structural approach to examine the determinants of loan amendments when lenders have the option to waive the covenant.

Our model distinguishes covenant waivers from renegotiations, a feature that prior theoretical work often overlooked (Chen and Wei [1993]).[8] Waivers are essentially costless, while renegotiations consume time and resources to construct a contract that all parties can agree to. Voting rules serve as a gatekeeper between the firm and the entire group of lenders—required lenders decide whether to waive the covenant or subject the firm to a renegotiation with all syndicate members. A novel feature that arises from the model is the tradeoff between the costs of false alarms (Gigler et al. [2009]) and efficiency losses from inefficient investments. The latter cost differs from the Type II errors documented in the prior literature, in which Bad states fail to trigger covenant violations due to either low accounting quality (Gigler et al. [2009]) or earnings manipulation (Magee and Sridhar [1996], Guttman and Marinovic [2018]). In our setting, agency conflicts within the lending syndicate may lead to suboptimal waivers, undermining the effectiveness of covenants. The optimal voting rule balances these two forces and improves contracting efficiency. Our results provide a rationale for the widespread use of voting rules in debt contracts. In addition, we derive and empirically test comparative statics from our model, shedding light on how lenders' incentives impact voting rules.

Second, this paper contributes to the growing literature examining coordination problems within lending syndicates by providing a theoretical framework to analyze how heterogeneous lender preferences affect syndicated loans. Agency costs and information asymmetry within the syndicate affect the lead lender's loan share (Sufi [2007]), loan spreads (Ivashina [2009]), the use of financial covenants (Saavedra [2018], Dass et al. [2020]), and loan renegotiations (Martin et al. [2024]). Accordingly, lenders form syndicates that minimize coordination costs (Champagne and Kryzanowski [2007], Li [2021], Bushman, Sohn, and Zhao [2025]). While several studies empirically examine contracting frictions within the lending syndicate, theoretical research on these frictions remains limited. One

[7] Bolton and Scharfstein's [1996] model assumes that voting rules affect the loan's payment terms. As we discuss in the institutional background, this assumption typically does not hold in practice, as most loans require unanimous approval for changes to the interest rate, payment schedule, or commitment amount (LSTA [2016]). For this reason, we build a model based on Gârleanu and Zwiebel [2009], which does not rely on such assumptions.

[8] The existing models often assume a costly renegotiation (e.g., Gârleanu and Zwiebel [2009]) or a straightforward liquidation (e.g., Gigler et al. [2009]) following the covenant violations. The only exception we are aware of is Luneva [2025]. She shows that most lending agreements have substantial incompleteness in the sense that contractable signals are noisy representations of the state. Accordingly, optimal investment behavior depends on lenders' and borrowers' ability to either waive covenants or renegotiate loan terms. We differ substantially from Luneva [2025] in that we examine how lenders set the optimal voting rule that determines whether lenders grant waivers.

related paper is Zhong [2021], who investigates coordination costs arising from the number of (homogenous) creditors involved in a contract.[9] Our paper differs from Zhong [2021] in that we analyze coordination costs stemming from heterogeneous lender incentives in waiving a covenant violation, which give rise to the need for voting rules.

Third, this paper contributes to the broader literature on incomplete contracting (for recent reviews, see Hart [2001], Roberts and Sufi [2009b], Armstrong, Guay, and Weber [2010], Christensen, Nikolaev, and Wittenberg-Moerman [2016]). These studies build on the idea that contracting parties such as borrowers and lenders cannot perfectly anticipate all future scenarios. This leads to potential hold-up problems that impede contracting efficiency (Aghion and Bolton [1992]). There is extensive literature examining how ex ante contract design, such as the use of covenants, can facilitate more efficient allocation of control rights and mitigate hold-up problems (e.g., Berlin and Mester [1992], Rajan and Winton [1995], Gigler et al. [2009], Gârleanu and Zwiebel [2009]). We add to this stream of literature by theoretically demonstrating how lenders use the required lenders clause, an important yet previously overlooked component of syndicated loan contracts, to efficiently allocate control in incomplete contracts.

The next section reviews institutional details on the required lenders clause. Sections 3 and 4 develop the model and predictions. Section 5 provides our empirical results. Section 6 concludes.

2. *Institutional Background*

The required lenders clause specifies the minimum fraction of loan holdings needed to modify a loan contract. In a typical syndicated loan, the lead lender represents the lending group when interacting with the borrower. The lead lenders perform due diligence, monitor the loan, and process principal and interest payments. However, lead lenders do not have sole discretion to amend contracts. Contract amendments require the consent of a minimum fraction of required lenders. Certain provisions in loan contracts are not subject to the required lenders clause. These exceptions, sometimes referred to as sacred rights, require unanimous consent of all lenders. Sacred rights are terms directly related to each lender's payoffs. They typically include changes to the interest rate, payment schedule, and commitment amount (Roberts and Sufi [2009b], LSTA [2016], Proskauer [2020], Whitney and Zelinsky [2017]). U.S. loan agreements typically require the consent of at least 51% of the outstanding principal to modify nonmaterial terms of the loan agreement.

[9] Zhong [2021] is similar to Bolton and Scharfstein [1996] in that they both examine the optimal syndicate size with lenders having homogeneous preferences, although Zhong examines a dynamic setting. The lenders in Zhong [2021] can accept or reject renegotiated amounts for their particular loan commitment, making the loan more like a set of homogeneous loans than a single syndicated loan package funded by multiple lenders.

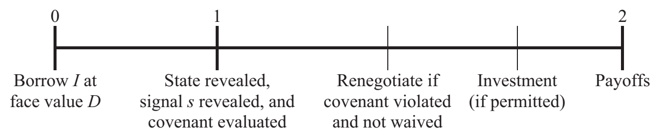

FIG. 1.—Timeline.
This figure plots the model's timeline.

A minimum of the required lenders must approve any other modifications, such as waivers to covenant violations or amendments to financial covenants.[10] Studies have shown that loan contracts are frequently amended to waive a covenant violation or amend a covenant to avoid technical default (Dichev and Skinner [2002], Roberts and Sufi [2009b]). The required lenders clause is a crucial feature in loan agreements as it directly affects the renegotiation process by allocating control rights, and the expectation of future loan renegotiations will be embedded in the ex ante contract design.[11] The required lenders clause is also used by rating agencies in evaluating loan contracts. For example, Moody's uses the amendment threshold as an input when constructing its Covenant Quality Indicator.[12]

3. Model

3.1 SETUP

This section develops a model where a firm obtains a loan from a syndicate of lenders with heterogeneous preferences for how to respond to a covenant violation. The structure follows Gârleanu and Zwiebel's [2009] single-lender model, where we introduce (1) an accounting-based covenant that, when violated, gives lenders veto power over the firm's interim investment decision, and (2) multiple lenders who vary in their preferences for the action to take following a covenant violation.

Figure 1 presents the model's timeline, where a firm borrows I to fund a project that yields a payoff of R. The loan's face value D ensures that all lenders at least break even, in expectation, and reflects contract features

[10] Appendix D provides an example of the required lenders clause. The credit agreement defines required lenders as 50% of the loan commitment. The agreement also states exceptions to the required lenders rule, including changes to loan commitments, payment schedules, the release of collateral, and the definition of required lenders. Any other modifications require approval by only the required lenders.

[11] While a lender could, in theory, accumulate voting power by acquiring additional loan shares in the secondary market, this is rare because (1) doing so requires taking on additional credit risk; (2) many syndicated loans are not actively traded—only about 10% of U.S. syndicated loans are traded (Bushman, Smith, and Wittenberg-Moerman [2010]); and (3) loan contracts often impose transfer restrictions or disqualified lender lists, which limit who can hold the debt. These restrictions are typically included in the credit agreement but are often redacted in public disclosures (Saavedra [2023]).

[12] We thank Moody's for providing an example of a loan quality report.

such as technical default conditions and voting thresholds, along with rational expectations about the borrower's and lenders' future actions. After borrowing, the firm can make an investment whose payoff depends on one of two states—Good (G) or Bad (B). We model the accounting-based covenant as a contractable signal that is informative about whether the state is Good (G) or Bad (B) for investment.

As in Gârleanu and Zwiebel [2009], the project has a positive NPV of y in the state G and a negative NPV of $-y$ in state B. In both states, the project increases (decreases) shareholders' (lenders') payoffs by x, which is a reduced-form representation of the impact of project risk on shareholder and lender payoffs.[13] The following are the monetary payoffs given the state, the investment choice, and the face value of debt D.

	Invest			
	Good (G)	Bad (B)	Not invest	
Shareholders	$R-D+x+y$	$R+x-D$	$R-D$	(1)
Debt	$D-x$	$D-x-y$	D	
Total	$R+y$	$R-y$	R	

This structure implies that in state G the project has positive value but lenders have postborrowing incentives to disallow it absent renegotiating the loan. Conversely, in state B, the project has negative value but shareholders have postborrowing incentives to allow it absent renegotiating the loan. We assume that renegotiation reduces lenders' payoffs by c_r, and that the project has a positive net-of-renegotiation-cost NPV: $\mathrm{P}(G)y > c_r$, where $\mathrm{P}(G)$ denotes the probability of the Good state.

A complete contract could achieve the first-best outcome of investing in the Good state and blocking investment in the Bad state. This yields a face value D_* that satisfies:

$$I = \mathrm{P}(G)(D-x) + (1-\mathrm{P}(G))\,D = D - \mathrm{P}(G)x \Rightarrow D_* = I + \mathrm{P}(G)x. \quad (2)$$

The shareholders' first-best expected payoff is then:

$$\mathrm{P}(G)\left(R + x + y - D_*\right) + (1-\mathrm{P}(G))\,(R - D_*) = R + \mathrm{P}(G)y - I. \quad (3)$$

We assume that the contract is incomplete in that all parties observe the state prior to executing the risky investment, but that this observation is not, itself, contractable. To allocate control over the investment decision to either lenders or shareholders, the lending agreement uses a contractable signal $s \in \{\underline{s}, \bar{s}\}$ that is informative about the state $\{G, B\}$. Formally, we assume the signal s satisfies the monotone likelihood ratio (MLR) property: $\frac{P(\bar{s}|G)}{P(\bar{s}|B)} > 1 > \frac{P(\underline{s}|G)}{P(\underline{s}|B)}$.

We assume that the contract precludes major investments when $s = \underline{s}$, which indicates a greater likelihood of state B. Because the signal reveals

[13] Lenders have a concave payoff function min{Face value, Firm payoff} so that risk reduces its expected value. Conversely, shareholders have a convex payoff function max{0, Firm value – Face value} so that risk increases its expected value.

information about the profitability of investment, we can interpret the covenant used in the loan as a performance covenant as in Christensen and Nikolaev [2012]. Lastly, the contract defines the Required Lenders as the minimum fraction of outstanding principal that can, among other things, waive covenant investment restrictions when $s = \underline{s}$, but cannot approve changes to any payment terms.

3.2 INVESTMENT AND WAIVER PREFERENCES

Absent renegotiation, shareholders prefer to invest regardless of the state, as implied by the payoffs in expression (1). Investment is optimal in state G, so that shareholder control yields no opportunities for renegotiation. Investment in state B gives shareholders the expected payoff $R + x - D$, and gives lenders the expected payoff of $D - y - x$. Shareholders can offer to forgo investment if lenders accept a reduced face value of $D' = D - y - x + c_r + \varepsilon$ for some small ε that we take to be negligible, which compensates lenders for their renegotiation costs and yields them the payoff $D' - c_r = D - y - x$. This yields the shareholders the payoff of $R - D' = R + x + y - c_r - D$.[14]

If the firm is in technical default ($s = \underline{s}$), then the Required Lenders have three possible courses of action:

1. Waive the covenant and allow the investment to proceed.
2. Deny a waiver, in which case the unaltered contract blocks investment.
3. Deny a waiver, but allow investment subject to renegotiating the lending agreement. Because renegotiation of a loan's payment terms requires unanimous consent from the lenders, this allows lenders to extract the value of the investment when renegotiating in state G.[15] Specifically, if lenders can block investment in state G, the shareholders receive a payoff of $R - D$. Lenders can renegotiate the loan in exchange for allowing investment, demanding a face value $D' = D + y + x > D$ that yields shareholders the payoff $R + y + x - D' = R - D$.[16] This yields lenders the net-of-renegotiation-costs monetary payoff $D' - x - c_r = D + y - c_r > D$.

The required lenders behave as gatekeepers between the firm and the entire group of lenders—they decide whether to waive the covenant or subject the firm to a renegotiation that will extract any surplus in state G.

[14] Alternatively, we can set the lenders' bargaining power to $\eta \in (0, 1)$ and the shareholders' bargaining power to $(1 - \eta)$. We could then maximize the Nash product to give shareholders (lenders) a postrenegotiation expected payoff of $R + x + (1 - \eta)(y - c_r) - D(D - x - y + \eta(y - c_r))$. The analysis remains largely unchanged because the debt's initial face value D accounts for any expected renegotiation.

[15] This may occur indirectly with lenders delegating the renegotiation to the lead lender, who must obtain new payment terms that satisfy all of the lenders (Diamond [1991]), or directly with all lenders participating in the renegotiations.

[16] One can think the higher face value reflecting a higher spread, early payment, reduction of borrowing base, or amendment fee as documented by Chen and Wei [1993].

If all lenders prefer the same course of action, the Required Lenders voting rule is arbitrary. We therefore assume that lenders vary in their willingness to waive covenants when $s = \underline{s}$. Specifically, we assume that some lenders realize a private benefit when the firm pursues the project without granting any concessions to lenders. These private benefits can reflect expectations of later business (Drucker and Puri [2005], Fernando et al. [2012]), relationships with management, and so on.

We denote lender n's share of the loan by μ_n and its private benefit by δ_n, and order the scaled private benefits $\tilde{\delta} = \delta/\mu$ so that $\tilde{\delta}_1 \geq \tilde{\delta}_2 \geq \ldots \geq \tilde{\delta}_N$, where N is the number of syndicate members. The following are the expected payoffs to a lender n who holds fraction μ_n of the loan and has private benefit δ_n from waiving the covenant:

State	Block	Renegotiate	Waive
G	$\mu_n D$	$\mu_n(D+y-c_r)$	$\mu_n(D-x)+\delta_n$
B	$\mu_n D$	N/A	$\mu_n(D-y-x)+\delta_n$

(4)

Accordingly, in state G, lender n prefers waivers to renegotiation only if its private benefits are sufficiently high:

$$\mu_n(D-x) + \delta_n > \mu_n(D+y-c_r) \Leftrightarrow \tilde{\delta}_n > \hat{\delta}_\ell = x + y - c_r\,. \tag{5}$$

When lenders control investment in state B, there are no gains from renegotiation and lender n prefers to waive the covenant and preserve the shareholder payoffs only if its private benefits are sufficiently high:

$$\mu_n(D-y-x) + \delta_n > \mu_n D \Leftrightarrow \tilde{\delta}_n > \hat{\delta}_h = x + y. \tag{6}$$

We refer to lenders who never waive covenants as having Low private benefits ($\tilde{\delta}_n \leq \hat{\delta}_\ell$), lenders who waive in only state G as having Moderate private benefits ($\tilde{\delta}_n \in (\hat{\delta}_\ell, \hat{\delta}_h)$), and lenders who always waive covenants as having High private benefits ($\tilde{\delta}_n \geq \hat{\delta}_h$). We summarize lenders' preferred actions by their private benefits in the table below:

	Lender's preferred action		
	$\tilde{\delta}_n \leq \hat{\delta}_\ell$ (Low private benefits)	$\tilde{\delta}_n \in (\hat{\delta}_\ell, \hat{\delta}_h)$ (Moderate private benefits)	$\tilde{\delta}_n \geq \hat{\delta}_h$ (High private benefits)
G	Renegotiate	Waive	Waive
B	Deny	Deny	Waive

(7)

If the loan agreement's required lenders clause requires fraction $\hat{\mu} \geq \frac{1}{2}$ of the loan commitments to approve a waiver, then the marginal vote will be held by the lender $n \leq N$ such that:

$$\mu_1 + \cdots + \mu_{n-1} < \hat{\mu} \leq \mu_1 + \cdots + \mu_{n-1} + \mu_n. \tag{8}$$

The marginal vote then depends on that lender's private benefits $\tilde{\delta}_n$. If lender n prefers to waive a covenant, so does any lender with private benefits greater than $\tilde{\delta}_n$ so that lender n's preference dictates whether the covenant will be waived.

3.3 VOTING RULE AND THE CHOICE OF THE MARGINAL VOTER

The optimal voting rule depends on the private benefits of the lender with the marginal vote. As discussed in the previous section, lenders' waiver preferences (i.e., High vs. Low private benefit) depend on two key levels. Lenders with private benefits $\tilde{\delta} < \hat{\delta}_\ell$ never waive covenants, and lenders with private benefits $\tilde{\delta} > \hat{\delta}_h$ always waive covenants, even in the Bad state. Given a voting threshold $\hat{\mu}$, we then denote by $\pi_\ell(\hat{\mu})$ the probability that the marginal vote has Low private benefits, and $\pi_h(\hat{\mu})$ the probability that the marginal vote has High private benefits. Stricter voting thresholds (higher $\hat{\mu}$) have a higher (lower) probability of the marginal vote having Low (High) private benefits. If there is no uncertainty about lenders' private benefits, then $\pi_\ell(\hat{\mu})$ and $\pi_h(\hat{\mu})$ are either zero or one for any voting threshold $\hat{\mu}$.

We assume that contracting parties have some uncertainty about lenders' private benefits.[17] Because of this, they cannot be sure of the marginal voter's private benefits and $0 < \pi_\ell(\hat{\mu}), \pi_h(\hat{\mu}) < 1$. To model $\pi_\ell(\hat{\mu})$ and $\pi_h(\hat{\mu})$, we denote by $F_\omega(\hat{\delta})$ the fraction of the lending syndicate with scaled private benefits less than or equal to $\hat{\delta}$, where the parameter ω indexes the distributions $F_\omega(\hat{\delta})$ in the sense of MLR dominance. This implies that $F_\omega(\delta)$ decreases in ω or, equivalently, for any given $\hat{\delta}$ and $\omega' > \omega$, $F_{\omega'}(\hat{\delta}) < F_\omega(\hat{\delta})$. As ω increases, the distribution $F_\omega(\hat{\delta})$ has fewer (more) lenders with scaled private benefits less than or equal to (greater than) $\hat{\delta}$. In other words, lenders' private benefits are increasing in ω.

We use the parameter ω to model uncertainty about lenders' private benefits. Specifically, we assume that the borrower and lenders have a common belief about the distribution of ω that is represented by the distribution function $G(\omega)$.[18] Uncertainty about ω implies that for each $\hat{\delta}$ the fraction $F_\omega(\hat{\delta})$ is a random variable. Because $\hat{\delta}_h > \hat{\delta}_\ell$, $F_\omega(\hat{\delta}_h)$ first-order stochastic dominates $F_\omega(\hat{\delta}_\ell)$.[19] To facilitate analysis, we make the stronger assumption that $F_\omega(\hat{\delta}_h)$ dominates $F_\omega(\hat{\delta}_\ell)$ in terms of the MLR ordering.

The private benefits of the marginal voter for threshold $\hat{\mu}$ corresponds to the $\tilde{\delta} = \hat{\delta}$ such that $1 - F_\omega(\hat{\delta}) = \hat{\mu}$ or, equivalently, fraction $\hat{\mu}$ of lenders have private benefits at least as high as $\hat{\delta}$. Because lenders' willingness to

[17] In practice, contracting parties often do not know the exact distribution of lenders' private benefits. The main intuition holds absent uncertainty about lenders' private benefits except that, as we illustrate later, there will typically be a range of optimal voting rules.

[18] The common beliefs assumption allows us to bypass the need for the borrower and lenders to extract private information via, for example, offering menus of contracts.

[19] Recall that ω orders the distributions $F_\omega(\hat{\delta})$ in terms of the MLR so that for fixed $\hat{\delta}$, $F_\omega(\hat{\delta})$ decreases in ω and the ω that solves $F_\omega(\hat{\delta}_h) = \tau$ is greater than the ω that solves $F_\omega(\hat{\delta}_\ell) = \tau$. This implies that for any $\tau \in (0, 1)$, $\mathrm{P}(F_\omega(\hat{\delta}_h) \leq \tau) \leq \mathrm{P}(F_\omega(\hat{\delta}_\ell) \leq \tau)$ with the inequality strict for some τ.

waive covenants increases with private benefits, the probabilities $\pi_\ell(\hat{\mu})$ and $\pi_h(\hat{\mu})$ that the marginal voter has Low or High private benefits are:

$$\begin{aligned}\pi_\ell(\hat{\mu}) &= \mathrm{P}(\text{Less than}\,\hat{\mu}\,\text{have private benefits greater than } \hat{\delta}_\ell) \\ &= \mathrm{P}\left(F_\omega(\hat{\delta}_\ell) \geq 1-\hat{\mu}\right), \\ \pi_h(\hat{\mu}) &= \mathrm{P}(\text{At least}\,\hat{\mu}\,\text{have private benefits greater than } \hat{\delta}_h) \\ &= \mathrm{P}\left(F_\omega(\hat{\delta}_h) \leq 1-\hat{\mu})\right).\end{aligned} \tag{9}$$

The probability that the marginal vote has Moderate private benefits is then $1-\pi_\ell(\hat{\mu})-\pi_h(\hat{\mu})$. Note that π_ℓ increases in $\hat{\mu}$ while π_h decreases in $\hat{\mu}$, reflecting that higher voting thresholds are more (less) likely to give the marginal vote to lenders with Low (High) private benefits.

To illustrate, figure 2 plots a specific example distribution of lenders' private benefits. The figure parameterizes $F_\omega(\hat{\delta})$ assuming that for all ω, 10% of lenders have zero private benefits so that a Zero-private-benefit lender is certain to be the marginal voter for any voting threshold of 90% or greater. We represent the fraction for the remaining 90% of lenders using a lognormal distribution with $F_\omega(\hat{\delta}|\hat{\delta}>0)=\Phi(\frac{\log\hat{\delta}-\omega}{\sigma})$ and the beliefs about ω using $G(\omega)=\Phi(\frac{\omega-\bar{\omega}}{\sigma_\omega})$ where $\Phi(\cdot)$ denotes the standard normal distribution (we refer to these assumptions as the lognormal/normal setting hereafter).[20] The blue line shows the expected distribution of the fraction of lenders with benefits at least as high as $\hat{\delta}$, that is, $1-\mathrm{E}[F_\omega(\hat{\delta})]$. Panel A illustrates uncertainty about the distribution of private benefits with the dark (light) shaded regions representing the 25th to 75th (10th to 90th) percentiles of beliefs about the distribution.

The 50th percentile of the expected distribution—where the blue line crosses the horizontal 50% line in figure 2, panel A—is 1.1, which lies between the example's $\hat{\delta}_\ell=1.0$ and $\hat{\delta}_h=1.5$. If there is no uncertainty about private benefits and all parties are certain that $\mathrm{E}[F_\omega(\hat{\delta})]$ represented the actual fractions of lenders' private benefits, any voting threshold between 50% and about 53.5% would guarantee that a Moderate-private-benefit lender has the marginal vote. Because Moderate-benefit lenders waive covenants in the Good state and enforce them in the Bad state, this eliminates inefficiencies in the event of a technical default. In fact, the contract could be further improved by eliminating the financial covenant and giving lenders unconditional veto rights over investments. This would achieve first-best outcomes by eliminating shareholder-initiated renegotiations when the covenant is not triggered in the Bad state.

However, when contracting parties are uncertain of the distribution of lender's private benefits (i.e., $F_\omega(\hat{\delta})$ is not known with certainty), a voting threshold of 50% in the above example will not guarantee the marginal voter being a Moderate-private-benefit lender. To see this, figure 2, panel B, illustrates the uncertainty as a density, and shows the probabilities $\pi_\ell(\hat{\mu})$ and $\pi_h(\hat{\mu})$ of the marginal vote having Low or High private benefits at

[20] We present derivations in appendix A.

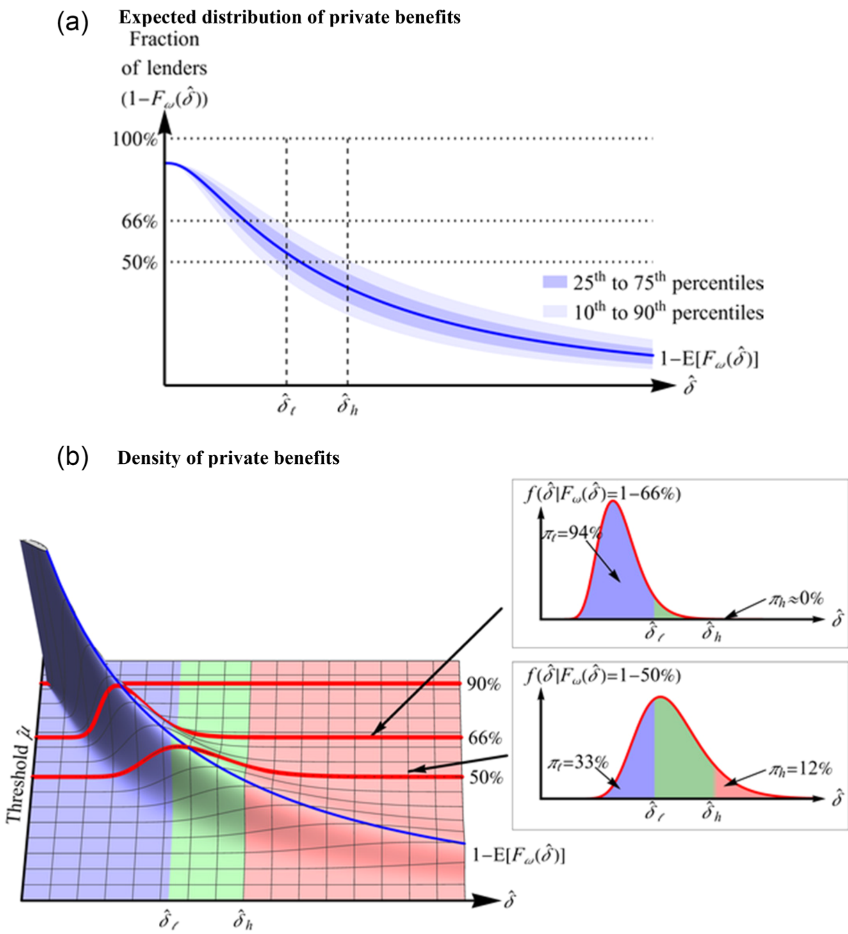

FIG. 2.—Distribution of lenders' private benefits.
This figure plots an example distribution of lenders' scaled private benefits $\tilde{\delta}$. In both panels, the blue line is the expected fraction of lenders with private benefits of at least $\hat{\delta}$, i.e., $(1 - \mathrm{E}[F_\omega(\hat{\delta})])$. In panel A, the dark (light) shaded areas reflect uncertainty via the 25th to 75th (10th to 90th) percentile range of beliefs about lenders' private benefits. Panel B plots the corresponding density where blue (red) regions denote lenders with Low (High) private benefits, and the inserts show the π_h and π_ℓ values for voting thresholds of 50% and 66%. The plots assume that 10% of lenders have zero private benefits and use $\mathrm{P}(\tilde{\delta} \geq \delta|\omega) = \Phi(\omega - \log\delta)$ (i.e., $\log\tilde{\delta} \sim \mathcal{N}(\omega, 1)$), $\omega \sim \mathcal{N}(0.25, 0.25^2)$ for the 90% of lenders with nonzero private benefits. The plots also use $y = 1$ and $x = c_r = 0.5$ so that $\hat{\delta}_h = 1.5$ and $\hat{\delta}_\ell = 1.0$.

the example thresholds $\hat{\mu} = 50\%$ and 66%. When the threshold is set at 50%, there is a 33% (12%) probability that the marginal voter has Low (High) private benefits, which can cause efficiency losses from unnecessary renegotiations in the Good state (excessive waivers in the Bad state). When the threshold is increased to a more stringent 66%, the probability that the marginal voter has Low (High) private benefits increases to 94%

(decreases to approximately 0%). In other words, higher voting thresholds reduce excessive waivers in the Bad state but increase the cost of unnecessary renegotiations in the Good state. The optimal voting rule balances these two forces.

3.4 SHAREHOLDER PAYOFFS

To determine shareholders' payoffs, we first walk through the derivation of the loan's face value D for a given voting rule $\hat{\mu}$. The debt's face value D must compensate each lender for its share μ_n of the loan proceeds I. In particular, the face value must satisfy the breakeven condition for the lender with the lowest scaled private benefits $\tilde{\delta}_N$, giving the breakeven condition after dividing through by μ_N:

$$
\begin{aligned}
I = {} & \underbrace{\mathrm{P}(\bar{s}, G)\,(D - x + \tilde{\delta}_N)}_{\text{State } G\text{, no violation}} + \underbrace{\mathrm{P}(\underline{s}, G)\,(D + \overbrace{(1-\pi_\ell)\,(\tilde{\delta}_N - x)}^{\text{Waive covenant}} + \overbrace{\pi_\ell\,(y - c_r)}^{\text{Renegotiate}})}_{\text{State } G\text{, violation}} \\
& + \underbrace{\mathrm{P}(\bar{s}, B)\,(D - x - y + \tilde{\delta}_N)}_{\text{State } B\text{, no violation}} + \underbrace{\mathrm{P}(\underline{s}, B)\,(D - \overbrace{\pi_h\,(x + y - \tilde{\delta}_N)}^{\text{Waive covenant}})}_{\text{State } B\text{, violation}}, \\
= {} & D - \underbrace{\big(\mathrm{P}(\bar{s}) + \mathrm{P}(\underline{s}, G)\,(1-\pi_\ell) + \mathrm{P}(\underline{s}, B)\pi_h\big)}_{\mathrm{P}(\text{Shareholder control or waive covenants})}(x - \tilde{\delta}_N) \\
& + \underbrace{\mathrm{P}(\underline{s}, G)\pi_\ell\,(y - c_r)}_{\text{Renegotiation gains}} - \underbrace{\big(\mathrm{P}(\bar{s}, B) + \mathrm{P}(\underline{s}, B)\pi_h\big)}_{\mathrm{P}(\text{State } B\text{, no violation or waived covenants})} y. \qquad (10)
\end{aligned}
$$

The face value D solves expression (10). The lender with the lowest scaled private benefits $\tilde{\delta}_N$ breaks even in expectation, while any lender n with higher private benefits earns an expected surplus proportional to $\tilde{\delta}_n - \tilde{\delta}_N$. Because shareholders can use a low face value to extract the value of lenders' private benefits, lenders benefit from including syndicate members with zero private benefits $\tilde{\delta}_N = 0$. For the remainder of the analysis, we assume that the syndicate includes at least one member with zero private benefits, which then determines the face value of debt and ensures that some lenders have Low private benefits.

After substituting the face value from (10), the shareholders' ex ante payoff is:

$$
\underbrace{R + \mathrm{P}(G)y - I}_{\text{First-best payoff}} - \underbrace{\big(\mathrm{P}(G|\underline{s})\pi_\ell c_r + \mathrm{P}(B|\underline{s})\pi_h y\big)}_{\text{Inefficiencies from covenant violation}} \mathrm{P}(\underline{s}) - \underbrace{\mathrm{P}(\bar{s}, B)\,c_r}_{\substack{\text{Inefficiencies from}\\ \text{shareholder}\\ \text{renegotiations}}}, \qquad (11)
$$

where the probabilities π_ℓ and π_h depend on the voting threshold $\hat{\mu}$ as described in the previous subsection. Compared to the first-best payoff, the inability to directly contract on the state (incomplete contract) results in efficiency losses from renegotiation costs and excessive investments in Bad state. Specifically, the $\mathrm{P}(G|\underline{s})\pi_\ell c_r$ term reflects the cost of what Gigler et al. [2009] describe as false alarms or Type I errors where the covenant gives lenders control rights in the Good state. When the marginal vote has Low private benefits, lenders will renegotiate rather than waive the covenant. The $\mathrm{P}(\bar{s}, B)c_r$ term reflects costs that stem from what Gigler et al. [2009] refer to as Type II errors where the Bad state fails to trigger a technical default (i.e., $s = \bar{s}$ in the Bad state). In such cases, shareholders control the investment decision and can demand a renegotiation where lenders accept a lower spread in exchange for shareholders forgoing investment (see section 3.2). The $\mathrm{P}(B|\underline{s})\pi_h y$ term is not present in Gigler et al. [2009] because their model does not consider lenders' private benefits from accommodating the borrower. It captures the cost of excessive investments in the Bad state, as a too-lenient voting rule assigns the marginal vote to a High Private Benefit lender that fails to block investments in the Bad state.

3.5 OPTIMAL VOTING RULE

The optimal voting rule minimizes the postviolation expected costs $\mathrm{P}(G|\underline{s})\pi_\ell c_r$ from false alarms plus the expected costs $\mathrm{P}(B|\underline{s})\pi_h y$ from lender control when the marginal voter has High private benefits and waives covenants in the Bad state. The voting rule does not impact the expected costs $\mathrm{P}(\bar{s}, B)c_r$ from shareholder control; rather, these costs depend solely on the informational properties of the signal $s \in \{\underline{s}, \bar{s}\}$.

Figure 3 illustrates how the voting rule impacts costs for the example from figure 2. In the example, the lowest possible threshold $\hat{\mu} = 50\%$ has a 12% chance of the marginal voter having High private benefits that result in waivers in state B and a 33% chance of the marginal voter having Low private benefits that result in renegotiations after false alarms. A $\hat{\mu} = 66\%$ voting rule nearly eliminates the chance of the marginal voter having High private benefits, but has a 94% chance of the marginal voter having Low private benefits and forcing renegotiations after false alarms. The optimal voting rule—about 55% in this example—balances the costs of renegotiation after false alarms against the costs of inefficient waivers when the covenant is triggered in state B and investment should be blocked.

The optimal voting rule $\hat{\mu}$ minimizes inefficiency costs, where $\hat{\mu}$ must weakly exceed $\frac{1}{2}$ to reflect majority preference for the binary choice of whether to waive the covenant:

$$\min_{\hat{\mu}\geq\frac{1}{2}} \mathrm{P}(G|s = \underline{s},)\pi_\ell(\hat{\mu})c_r - \mathrm{P}(B|s = \underline{s})\pi_h(\hat{\mu})y. \tag{12}$$

The following result, proved in the appendix, specifies the optimal voting rule:

Proposition 1. *The optimal loan contract sets the voting rule at the maximum of* $\frac{1}{2}$ *or the unique* $\hat{\mu}$ *that solves* $-\frac{\pi'_h(\mu)}{\pi'_\ell(\mu)} = \frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}$.

The optimal voting rule balances the marginal reduction in the costs of covenant waivers in state B $(\pi'_h(\mu)\mathrm{P}(B|\underline{s})y < 0)$ against the marginal

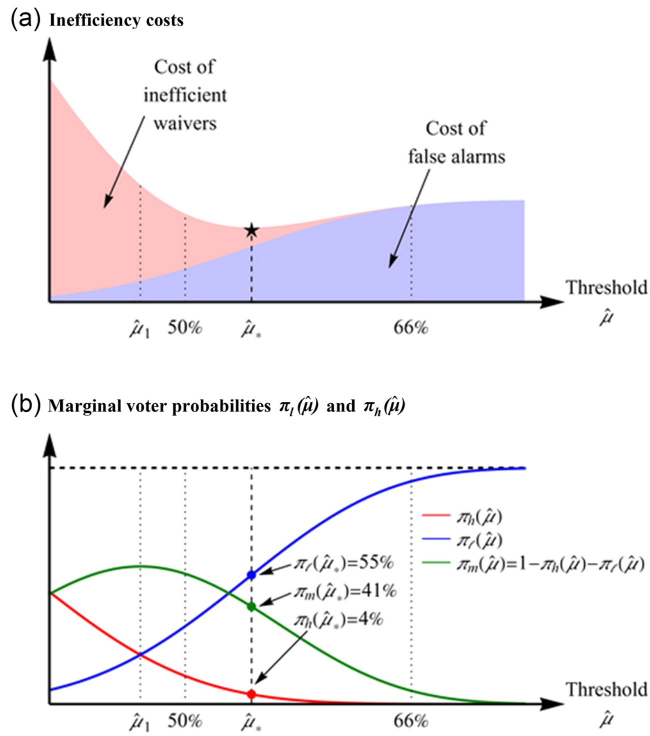

FIG. 3.—Optimal voting threshold.
This figure plots the total inefficiency costs from covenant violations (panel A) and the probabilities $\pi_\ell(\hat{\mu})$ ($\pi_h(\hat{\mu})$) of the marginal voter having Low (High) private benefits (panel B) as a functions of the voting threshold $\hat{\mu}$. The threshold $\hat{\mu}_1$ maximizes the probability of the marginal voter having Moderate private benefits, and the threshold $\hat{\mu}_*$ minimizes the expected inefficiency costs. The expected inefficiency costs are $P(G|\underline{s})c_r \times \pi_\ell(\hat{\mu})$ from false alarms plus $P(B|\underline{s})y \times \pi_h(\hat{\mu})$ from inefficient waivers. The plots use the example parameters from figure 2.

increase in the costs of renegotiations in state G $(\pi'_\ell(\mu)P(G|\underline{s})c_r)$. The optimal voting threshold is about 55% for the example in figure 3. Note that the optimal voting threshold does not maximize the probability that the marginal vote has Moderate private benefits at the peak of the green curve, which corresponds to the $\hat{\mu}$ that solves $-\frac{\pi'_h(\mu)}{\pi'_\ell(\mu)} = 1$. The optimal voting threshold tends to be stricter than that. To see why, this is because we assume that renegotiation costs are small relative to the cost of implementing a bad project ($c_r < P(G)y$), which we expect often to be the case in practice. This implies that $P(G|\underline{s}) < P(B|\underline{s})$ or, equivalently, $P(G|\underline{s}) < \frac{1}{2}$, is sufficient for $\frac{P(G|\underline{s})}{P(B|\underline{s})}\frac{c_r}{y} < 1$.[21]

[21] Formally, $\frac{P(G|\underline{s})}{P(B|\underline{s})}\frac{c_r}{y} < 1$ if and only if $P(G) < 1/(1 + \frac{P(\underline{s}|G)}{P(\underline{s}|B)}\frac{c_r}{y})$. The upper bound ranges from $1/(1 + \frac{c_r}{y})$ for an uninformative signal ($P(\underline{s}|B) = P(\underline{s}|G)$) to 1 for a perfectly informative signal ($P(\underline{s}|B) = P(\bar{s}|G) = 1$).

Borrowers always have the option to eschew covenants and accept the corresponding adjustment to the loan's face value. If the contract has no covenants, then shareholders will offer to renegotiate in the Bad state, yielding expected costs of $\mathrm{P}(B)\,c_r$. As shown in expression (11), a contract with covenants reduces the costs of shareholder-initiated renegotiations to $\mathrm{P}(B, \bar{s})\,c_r$ at the expense of potential lender-initiated renegotiations in the Good state and potential covenant waivers in the Bad state. The following corollary states that if the precision of the signal $s \in \{\underline{s}, \bar{s}\}$ is sufficiently high and/or the probability $\mathrm{P}(G)$ of the Good state is sufficiently low such that $\frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})} < 1$, then a contract with covenant-based contingent control dominates a contract with no covenants.

Corollary 1.1. *If* $\frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})} < 1$, *then the optimal loan contract with covenants yields lower inefficiency costs than a contract without covenants.*

4. *Comparative Statics and Empirical Predictions*

In this section, we derive comparative statics from our model. We then provide several testable predictions regarding the relationship between voting rules and lenders' private benefits, conservatism and accounting-based covenants.

4.1 CHANGES IN PRIVATE BENEFITS

Our first set of comparative statics pertains to a shift in the distribution of lenders' private benefits. We model a shift in private benefits as a shift in the distribution $G(\omega)$ of the parameter ω in the sense of the MLR ordering.[22] As we discuss in section 3.3, higher values of ω correspond to distributions $F_\omega(\hat{\delta})$ with a greater fraction of lenders having high private benefits. An upward shift in the distribution $G(\omega)$ corresponds to contracting parties believing that the high private benefit (ω) distributions are more likely. In the context of our parametric example with lognormal/normal distribution, an MLR shift corresponds to an increase in the mean $\bar{\omega}$ of ω.

The following proposition, proved in the appendix, indicates that when the contracting parties believe that lenders are more likely to have High private benefits, they prefer a stricter threshold to avoid lenders waiving covenants in the Bad state. Conversely, when they believe lenders are more likely to have Low private benefits, they prefer a more lenient threshold that avoids costly renegotiations in the event of false alarms.

Proposition 2. *The optimal voting rule weakly increases with higher levels of private benefits (MLR dominance shift in the distribution $G(\omega)$).*

Figure 4 illustrates a shift in private benefits using the example in figure 2. The optimal voting rule for the example in figure 2 is $\hat{\mu} = 55\%$.

[22] We assume that the changes do not affect the conditional distribution of $\tilde{\delta}$ given ω.

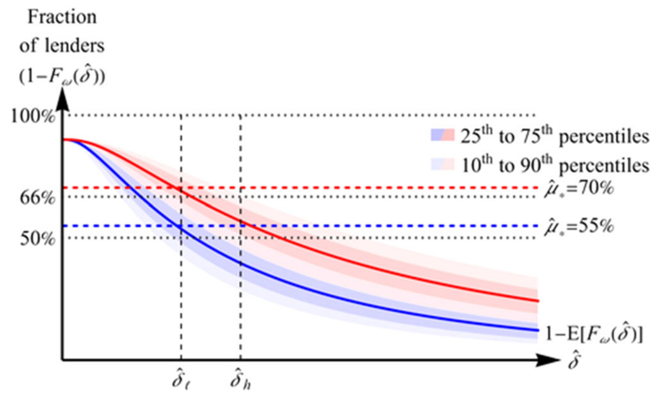

FIG. 4.—Example voting thresholds.
This figure illustrates the optimal voting threshold, and that the threshold increases with increases in the contracting parties' beliefs about lenders having high private benefits (Proposition 2). Similar to figure 2, panel A, the solid lines plot the expected distribution of private benefits $(1 - \mathrm{E}[F_\omega(\hat{\delta})])$, and the dark-shaded (light-shaded) areas plot the 25th to 75th (10th to 90th) percentiles. The blue line and areas correspond to the example in figure 2, with an optimal voting rule of $\hat{\mu} = 55\%$. The red line and areas correspond to a shift in beliefs from $\bar{\omega} = 0.25$ to $\bar{\omega} = 0.75$, with the new optimal voting rule of $\hat{\mu} = 70\%$.

In the parametric example, an MLR shift in the distribution of ω corresponds to an increase in its mean $\bar{\omega}$, which we denote by the red line and red-shaded areas. Under the shifted beliefs, the baseline $\hat{\mu} = 55\%$ voting threshold yields a 60% (3%) probability that the marginal voter has High (Low) private benefits. The optimal threshold for the shifted distribution is 70%, which yields a 4% (55%) probability that the marginal voter has High (Low) private benefits.

The distribution of private benefits impacts the face value D and shareholder payoffs via only the probabilities π_ℓ and π_h (see expression (10)). Absent any changes to voting rules, expression (9) implies that an increase in lenders' private benefits increases π_h and decreases π_ℓ. This leads to a higher face value D and lower shareholder payoffs. The higher voting threshold mitigates the negative impact on the face value and shareholder payoffs. In the lognormal/normal parameterization of the model, the higher voting threshold exactly offsets the impact that an MLR shift in the distribution G has on π_ℓ and π_h, so that the face value and expected shareholder payoffs are unchanged (see figure 4). In other words, in this example, the voting threshold nullifies the impact of changes in the levels of private benefits. However, we note that this complete offset is not a general result.

In addition to shifts in the level of private benefits, we also examine shifts in the uncertainty about lenders' private benefits. The following remark, proved in appendix A, summarizes the impact of greater uncertainty about private benefits (higher variance σ_ω^2), where we impose the lognormal/normal distribution assumption to simplify the analyses:

Remark: In the lognormal/normal setting with $\frac{P(G|\underline{s})}{P(B|\underline{s})}\frac{c_r}{y} < 1$, uncertainty about private benefits (σ_ω^2) has the following impacts:

a. The optimal voting threshold $\hat{\mu}_*$ increases with σ_ω^2.
b. The probability π_ℓ increases with σ_ω^2. There is a $\bar{\sigma}_\omega^2 > 0$ such that π_{h} increases (decreases) with σ_ω^2 for $\sigma_\omega^2 < \bar{\sigma}_\omega^2$ ($\sigma_\omega^2 > \bar{\sigma}_\omega^2$).
c. The shareholders' ex ante payoff decreases with σ_ω^2.

Under the condition $\frac{P(G|\underline{s})}{P(B|\underline{s})}\frac{c_r}{y} < 1$, greater uncertainty results in a higher voting threshold because expected efficiency losses from waiving a covenant in a Bad state (i.e., marginal voter having High-private benefit) are larger than the expected renegotiation costs after a false alarm (i.e., marginal voter having Low-private benefit). The optimal voting rule increases with uncertainty as a way to trade off the expected costs of waivers in the Bad state against the relatively lower costs of renegotiations in the Good state. The resulting probability π_ℓ of a Low-private-benefit lender having the marginal vote increases, while the probability π_h of a High-private-benefit lender may increase or decrease depending on parameters.[23] The shareholders' expected payoff always decreases with σ_ω^2 because the change in the voting rule cannot fully compensate for the impact of changing uncertainty. Part c of the Remark provides a theoretical motivation for empirical findings that lead banks prefer to form syndicates with lenders that are geographically closer (Sufi [2007]) or with whom the lead bank has prior syndicate relationships (Champagne and Kryzanowski [2007]), both of which may correspond to lower uncertainty about lender preferences.[24]

4.2 ACCOUNTING CONSERVATISM, THE NUMBER OF COVENANTS, AND VOTING THRESHOLDS

In this subsection, we extend our analyses to shed light on how companies' accounting systems affect the voting rules to waive covenants. We first analyze accounting conservatism. Early studies argue that conservative accounting is useful in debt contracting because it facilitates the transfer of control rights in Bad states (e.g., Watts [2003]). However, Gigler et al. [2009] show that conservative accounting might impede contract efficiency due to false alarms.

[23] As shown in the appendix, the direct effect of σ_ω^2 on π_ℓ is positive if and only if $\sigma_\omega^2 < \bar{\sigma}_\omega^2$. The direct effect on π_h is always positive. The indirect effect on π_ℓ (π_h) via changes in $\hat{\mu}_*$ is positive (negative).

[24] In the online appendix, we analyze the impact of σ_ω^2 on the face value D, which is related to the cost of borrowing, that is, the spread of roughly $D/I - 1$. Intuitively, one might expect that the debt's face value D increases with uncertainty. We show that this is not always the case and may not even be typical.

To examine how accounting conservatism affects the optimal voting rule, we impose further structure from Caskey and Laux [2017]. We implicitly assume that accounting policies are stickier than specific lending agreements. Specifically, we denote accounting conservatism by a parameter $c \in [\underline{c}, \bar{c}]$ where a higher c reflects a higher degree of conservatism in the sense that conservatism increases the likelihood of observing the low signal $\underline{s}$, regardless of the state, and increases the likelihood ratio $\frac{\mathrm{P}(s|G)}{\mathrm{P}(s|B)}$, regardless of the signal. We further assume that $\frac{\mathrm{dP}(\underline{s}|B;c)}{\mathrm{d}c} = \gamma \frac{\mathrm{dP}(\underline{s}|G;c)}{\mathrm{d}c} > 0$ or, equivalently, $\frac{\mathrm{dP}(\bar{s}|B;c)}{\mathrm{d}c} = -\gamma \frac{\mathrm{dP}(\underline{s}|G;c)}{\mathrm{d}c} < 0$, where the parameter $\gamma > 0$ allows conservatism to have a different impact on state B versus G.[25] Conservative accounting increases the likelihood of realizing the low signal $\underline{s}$. When $\gamma > 1$, more conservative accounting reduces false positives $\mathrm{P}(\bar{s}|B; c)$ more than it increases false negatives $\mathrm{P}(\underline{s}|G; c)$, and vice versa when $\gamma < 1$. The following proposition summarizes how a firm's accounting conservatism relates to the voting rule in its lending agreement:

Proposition 3. *The optimal voting rule is weakly decreasing in the degree c of accounting conservatism.*

The intuition for Proposition 3 follows from Gigler et al.'s [2009] prediction that, in a single-lender setting where a future project's expected payoff exceeds its abandonment value, conservative accounting detracts from contract efficiency. As in their setting, when lenders with Low private benefits have the marginal vote, false alarms lead to costly renegotiation. Because conservative accounting increases the likelihood of false alarms, the voting rule compensates by lowering the voting threshold for waivers.

Companies can tailor accounting-based covenants in debt contracts by utilizing multiple metrics, which we model as the signal s (see section 3.1). To examine how the number of financial covenants impacts the optimal voting threshold, we modify our model to allow contracts to use multiple, distinct signals $s_m \in \{\underline{s}_m, \bar{s}_m\}$ for $m \in \{1, 2, \ldots, M\}$. We assume that covenants follow the typical practice that investment restrictions follow the violation of any technical default threshold.[26] In other words, lenders control the investment decision unless $s_m = \bar{s}_m$ for $m = 1, 2, \ldots, M$, which we denote by the event $\bar{S}_M$. Proposition 1's definition for an interior voting threshold is then:

$$-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})} = \frac{1 - \mathrm{P}(\bar{S}_M|G)}{1 - \mathrm{P}(\bar{S}_M|B)} \frac{\mathrm{P}(G)}{\mathrm{P}(B)} \frac{c_r}{y}. \tag{13}$$

We further assume that each new signal is incrementally informative in the sense that state B is more likely than state G to trigger a default $(\mathrm{P}(\underline{s}_M|\bar{S}_{M-1}, B) > \mathrm{P}(\underline{s}_M|\bar{S}_{M-1}, G))$. The following proposition summarizes

[25] See Caskey and Laux [2017] for technical restrictions on the parameter γ.

[26] In practice, companies often use multiple covenants as trip wires for borrowers. Each covenant is assessed individually, such that the violation of any one could lead to either a renegotiation or a waiver (e.g., Dichev and Skinner [2002]; Demerjian and Owens [2016]).

how the voting threshold varies with the number of covenants in the lending agreement:

Proposition 4. *The voting threshold for contract with M covenants is weakly less than for a contract with M – 1 covenants if and only if there is a sufficiently high probability of the M^{th} covenant triggering a technical default in state G. If signals are IID, then this condition always holds.*

Proposition 4 states that lending agreements will loosen voting rules to counteract the addition of covenants that may trigger a false alarm by defaulting in state *G*. When signals are statistically identical, the voting rule always decreases with the number of covenants.

4.3 EMPIRICAL PREDICTIONS

We formulate our empirical predictions based on the comparative statics from our model. Our first prediction pertains to how private benefits impact the optimal voting rule. As noted in Proposition 2, the optimal voting rule increases with lenders' private benefits. When the syndicate includes more lenders with high private benefits who might waive covenants for bad projects, then the voting rule will be higher to ensure those lenders do not have the marginal vote. Conversely, when the syndicate includes more lenders with Low private benefits who might extract concessions when covenants yield false alarms, then the voting rule will be lower to ensure those lenders do not have the marginal vote. This gives us the first prediction:

Prediction 1: *Voting rules are more stringent (lenient) when more lenders have High (Low) private benefits in siding with the borrower.*

We use two empirical proxies for lenders' private benefits to test this prediction. First, prior studies find evidence that banks may be more lenient to borrowers (e.g., waive covenants) if they believe that will help them keep underwriting business (Drucker and Puri [2005], Fernando et al. [2012]). Therefore, we use past underwriting business as a proxy for syndicates with High private benefits lenders, and predict more stringent voting thresholds for those syndicates. Second, we use hedge fund participation in the syndicate as a proxy for Low private benefits, which reflects the notion that hedge funds are hard bargainers that are more difficult to renegotiate with (Jiang et al. [2012]). Accordingly, we expect voting rules to be more lenient when there are hedge fund participants in the syndicate.

Next, we examine the association between voting thresholds and the firm's accounting policies and accounting-based signals used in covenants. The model indicates that firms with more conservative accounting prefer lower voting thresholds (Proposition 3) and contracts with more covenants have lower voting thresholds (Proposition 4). We therefore predict a negative association between the voting rule and both accounting conservatism and the number of covenants. We follow Basu [1997] in measuring accounting conservatism. To assess the number of covenants, we focus on performance covenants because they better match our model's construct of a signal about future profitability (Christensen and Nikolaev [2012]).[27] This gives our second and third predictions:

Prediction 2: *Voting thresholds are negatively correlated with accounting conservatism.*

Prediction 3: *Voting thresholds are negatively correlated with the number of performance covenants*

5. Empirical Tests

5.1 TEST SPECIFICATION

To investigate the determinants of the amendment threshold, we employ the following regression framework for firm i, loan package ℓ, and year t:

$$\begin{aligned}\begin{matrix}\text{Voting}\\ \text{threshold}_{i\ell t}\end{matrix} &= b_0 + b_1 X_{i\ell t} + b_2 \begin{matrix}\text{Syndicate}\\ \text{relationship}\\ \text{intensity}_{i\ell t}\end{matrix} + b_3\#\text{Lenders}_{i\ell t} + b_4\text{Spread}_{i\ell t}\\ &\quad + b_5 \begin{matrix}\text{Loan}\\ \text{Amount}_{i\ell t}\end{matrix} + b_6\text{Maturity}_{i\ell t} + b_7\text{ROA}_{it} + b_8\text{Leverage}_{it}\\ &\quad + b_9 \log(\text{Assets}_{it}) + \gamma_i + \tau_t + \lambda_\ell + e_{i\ell t}.\end{aligned} \tag{14}$$

We use two measures of our outcome variable *Voting Threshold.* The first measure, *Amendment Threshold (Amount),* is the minimum percent of the total loan amount required to approve an amendment, which Dealscan reports as the variable required lenders.[28] The second measure, *Percent of Lenders Required,* is the minimum percent of the lenders required to approve the amendment. It is defined as the minimum number of lenders required to approve an amendment divided by the total number of lenders in the syndicate (Bolton and Scharfstein [1996]). We derive this measure from the *Amendment Threshold (Amount)* and each lender's share of the loan.[29] The *Percent of Lenders Required* better captures our theoretical construct for the relative reliance on arm's-length lenders to approve loan modifications

[27] There is another type of covenants called capital or balance sheet covenants, which are formulated using balance sheet information (Demerjian [2011]; Christensen and Nikolaev [2012]). Unlike performance covenants, capital covenants are often used to address the incentive misalignment between the borrower and the lenders (Christensen and Nikolaev [2012]). We discuss the implication of capital covenants on voting rules at the end of Section 5.4.

[28] We verify the accuracy of the Dealscan data by conducting a textual analysis that compares Dealscan's "required lender" variable to the information provided by EDGAR. This test confirms that the Dealscan data are consistent with the data extracted from the loan contracts that firms file with the SEC and post on EDGAR.

[29] For example, suppose that the *Amendment Threshold (Amount)* is 51%, and the syndicate includes three lenders where the lead arranger holds 40% of the loan, and the two participant banks each hold 30%. Then this *Percent of Lenders Required* equals 2/3 because any loan modification requires the agreement of at least two lenders, and the syndicate includes three members.

as it takes in account the information on each lender's loan allocation. However, the measure requires frequently unpopulated Dealscan data on individual syndicate members' holdings. Therefore, we also present results using the *Amendment Threshold (Amount)* dependent variable, which is available for a larger sample.

The independent variable $X_{i\ell t}$ varies based on our prediction. To test for a positive relation between voting thresholds and underwriting relationships, we use an indicator variable that equals one if at least one of the lenders has underwritten the borrower's security issuances in the past three years. We retrieve security issuance data from Securities Data Company's (SDC's) new issues database, which covers equity and bond issuances in the United States and provides information on underwriters for each security issuance. Following Jeon et al. [2015], we identify an underwriter as the lead underwriter if SDC classifies it as the bookrunner or joint bookrunner of the issuance. To test for a negative relation between voting thresholds and syndicates that include hedge fund lenders, we match Dealscan lenders to hedge funds in the Hedge Fund Research (HFR) database. We use an indicator variable that equals one if at least one of the participant banks is a hedge fund.[30]

To test our predictions that more lenient voting rules are associated with more conservative accounting and with more covenants, we use Basu's [1997] conservatism measure and the number of performance covenants (e.g., debt-to-EBITDA) in the contract, respectively. Performance covenants are those that use income statement or cash flow statement information (Christensen and Nikolaev [2012]). In addition, while we do not have a prediction with respect to capital covenants, that is, covenants that rely on balance sheet information, we nevertheless control for them in some of the regression specifications as robustness checks.

We control for the prior syndicate relationship between the lead lender and non–lead lenders (i.e., *Syndicate Relationship Intensity*). Empirical studies show that lead lenders mitigate within-syndicate conflicts by selecting participants with whom they have prior syndicate relationships (e.g., Champagne and Kryzanowski [2007], Li [2021]). Following Ivashina [2009] and Saavedra [2018], we measure the intensity of the prior syndicate relationship between the lead lender and non–lead lenders as the percentage of the lead lenders' deals over the past three years that include at least one of the current participants.

Our model indicates that a prior syndicate relationship can affect voting rules through two channels: (1) It increases the lenders' reputational costs from leniency, which behaves like a reduction in the private benefit from

[30] About 1.5% of the loans in our sample have a hedge fund participant. The most frequent hedge fund lenders in our sample are Eaton Vance Management, Oppenheimer, Octagon Credit Investors, CypressTree Investment Management, and Mountain Capital.

waiving covenants (Proposition 2);[31] (2) it reduces the uncertainty about lenders' private benefits, resulting in a more lenient voting rule (Remark). Both channels predict that the optimal voting rule decreases in the intensity of prior syndicate relationships between the lead lender and participants.

We also control syndicate size (i.e., # *Lenders*) in all of our analyses. Ex ante, the association between syndicate size and voting thresholds is not straightforward. On the one hand, a larger syndicate size is associated with higher renegotiation costs (e.g., Asquith, Beatty, and Weber [2005], Saavedra [2018]), which our model predicts will reduce the voting threshold.[32] On the other hand, larger syndicates tend to reduce each lender's share of the loan. This alters the distribution of scaled private benefits $\tilde{\delta}$ and can result in a higher or lower voting threshold.

In addition to the two aforementioned control variables, our tests also include various package- and firm-level controls. Specifically, the package-level controls are loan spread, amount, and maturity. The firm-level controls are total assets, ROA, and book leverage. All control variables are winsorized at 1% to 99% level. Following prior literature (Bharath et al. [2011], Prilmeier [2017]), we include fixed effects γ_i for the borrower's industry (two-digit SIC code) and τ_t for the loan starting year. In addition, we include lead lender fixed effects λ_ℓ to control for heterogeneity among lenders.[33]

5.2 SAMPLE CONSTRUCTION AND SUMMARY STATISTICS

Table 1 presents the sample construction. Our base sample consists of all Dealscan loans that can be linked to Compustat Quarterly Fundamental Data using the Dealscan-Compustat Linking Table for all U.S. companies. Our sample starts in 1995 and ends in 2017, which is the year the Dealscan-Compustat Linking Table ends. We exclude loans that are missing the required lenders variable in Dealscan. We also exclude loans with only one lender, as the amendment threshold does not apply to these loans.[34] Lastly, we require nonmissing values for loan- and firm-level controls. Our final sample consists of 17,568 loan packages. Table 2 reports the distribution of *Amendment Threshold (Amount)*. Approximately 75% of the loans set

[31] Champagne and Kryzanowski [2007] show that banks with past syndicate relationships are more likely to form syndicates in the future. If a bank expropriates other syndicate members by being overly lenient to the borrower for future private benefits, this would potentially incur significant reputational damage that jeopardize future syndications opportunities.

[32] The optimal voting threshold solves $-\frac{\pi_h'(\mu)}{\pi_\ell'(\mu)} = \frac{c_r}{y}$ where $-\frac{\pi_h'(\mu)}{\pi_\ell'(\mu)}$ decreases with μ so that a higher c_r leads to a smaller optimal threshold μ.

[33] We treat loans with more than one lead lender as one group when applying lender fixed effects. These loans account for 8.4% of our sample. Excluding these loans does not affect our main results.

[34] Many contracts specifically define required lenders as two or more lenders (i.e., see https://www.sec.gov/Archives/edgar/data/313616/000110465903022902/a03-3858_1ex10d1.htm).

TABLE 1
Sample Construction

	# Loans
# Loans in Dealscan	51,976
Minus	
Missing required lender data	(30,752)
Loans with only one lender	(2,389)
Missing loan/firm level controls	(1,267)
# Loans in Sample	17,568
Minus	
Loans with lender's share	(9,331)
# Loans in Subsample	8,237

This table provides details on the construction of our samples of Dealscan loan packages.

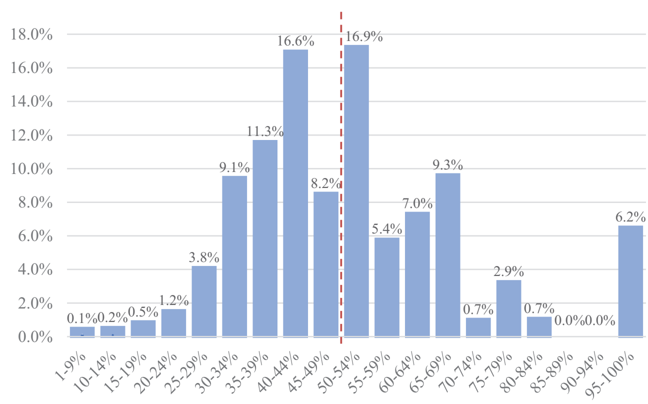

FIG. 5.—Distribution of the percent of lenders required.
This figure plots the distribution of *Percent of Lenders Required*, which is the minimal number of lenders required to approve the loan divided by the total number of lenders. The *X* axis is *Percent of Lenders Required*. The *Y* axis is the fraction of loans that fall into each bin. The red dashed line represents the 50% mark.

the amendment threshold to 51%. Twenty-one percent of the loans set this threshold to 66.7%. The remaining 4% of the loans have various thresholds between 51% and 100%.

As described in the previous subsection, our second measure of voting rules, *Percent of Lenders Required*, uses data on each lender's share of the loan. However, these data are missing for a large number of loans in Dealscan. Because of this, we create a subsample of 8,237 loan packages with non-missing lender shares. Figure 5 plots the distribution of *Percent of Lenders Required*. The average (median) percent of lenders required to approve an amendment is 50.4% (46.7%). Approximately half of the loans require at least 50% of lenders to approve an amendment. The pairwise correlation between our measures of voting rights (i.e., *Percent of Lenders Required* and *Amendment Threshold (Amount)*) is 0.61, suggesting that they capture similar constructs. In untabulated analysis, we separate the sample based on

TABLE 2
Summary Statistics

Panel A								
Loans with Lender's Holding	*N*	Mean	*SD*	p1	p25	p50	p75	p99
Percent of Lenders Required	8,237	0.504	0.179	0.200	0.385	0.467	0.600	1.000
Have Underwriting Relationship	8,237	0.153	0.360	0.000	0.000	0.000	0.000	1.000
Have Hedge Fund Participant	8,237	0.008	0.089	0.000	0.000	0.000	0.000	0.000
Syndicate Relationship Intensity	8,237	0.114	0.101	0.000	0.031	0.087	0.176	0.375
# Lenders	8,237	10.910	7.914	2.000	5.000	9.000	15.000	38.000
Book Leverage	8,237	0.317	0.214	0.000	0.165	0.302	0.436	1.010
ROA	8,237	0.009	0.027	−0.129	0.003	0.010	0.019	0.076
Log (Asset)	8,237	7.392	1.686	3.969	6.114	7.309	8.582	11.214
Log (Amount)	8,237	19.458	1.187	16.973	18.644	19.390	20.253	22.110
Log (Maturity)	8,237	3.696	0.607	1.792	3.584	3.989	4.094	4.554
Loan Spread	8,055	1.516	1.070	0.200	0.750	1.250	2.000	5.500
Conservatism	5,163	0.525	0.499	0.000	0.000	1.000	1.000	1.000
# Performance Covenants	7,539	1.463	0.978	0.000	1.000	1.000	2.000	4.000
# Capital Covenants	7,539	0.843	0.795	0.000	0.000	1.000	1.000	3.000

(*Continued*)

TABLE 2—*(Continued)*

All Loans	*N*	Mean	*SD*	p1	p25	p50	p75	p99
Percent of Lenders Required	17,568	0.551	0.074	0.510	0.510	0.510	0.600	0.750
Have Underwriting Relationship	17,568	0.156	0.363	0.000	0.000	0.000	0.000	1.000
Have Hedge Fund Participant	17,568	0.015	0.122	0.000	0.000	0.000	0.000	1.000
Syndicate Relationship Intensity	17,568	0.108	0.098	0.000	0.027	0.081	0.167	0.371
# Lenders	17,568	10.069	7.720	2.000	4.000	8.000	14.000	38.000
Book Leverage	17,568	0.340	0.232	0.000	0.174	0.318	0.466	1.135
ROA	17,568	0.007	0.030	−0.152	0.002	0.009	0.019	0.083
Log (Asset)	17,568	7.340	1.605	4.009	6.160	7.253	8.420	11.214
Log (Amount)	17,568	19.522	1.179	16.973	18.644	19.519	20.367	22.181
Log (Maturity)	17,568	3.779	0.598	1.792	3.584	4.094	4.094	4.605
Loan Spread	17,048	1.809	1.227	0.200	0.950	1.500	2.500	6.250
Conservatism	10,580	0.534	0.499	0.000	0.000	1.000	1.000	1.000
# Performance Covenants	15,985	1.639	1.012	0.000	1.000	2.000	2.000	4.000
# Capital Covenants	15,985	0.693	0.772	0.000	0.000	1.000	1.000	3.000

(Continued)

TABLE 2—*(Continued)*

Panel B. Amendment Threshold by Industry (Top 10)

Rank	SIC	Industry	# Loans	Percent Lender Required	Supermajority Percent[a]
1	49	Electric, Gas and Sanitary Services	842	47.9%	17.93%
2	67	Holding and Other Investment Offices	729	53.2%	49.73%
3	13	Oil and Gas Extraction	573	55.8%	49.82%
4	73	Business Services	540	52.4%	20.83%
5	35	Machinery and Computer Equipment	366	49.1%	23.51%
6	28	Chemicals and Allied Products	346	48.2%	14.78%
7	63	Insurance Carriers	302	48.4%	20.49%
8	36	Electronic & Other Electrical Equipment & Components	286	53.0%	24.52%
9	48	Communications	266	46.0%	13.51%
10	38	Measuring, Photographic, Medical, & Optical Goods	258	50.8%	14.71%
		Total # Loans	8,237		

(*Continued*)

TABLE 2—*(Continued)*

Panel C. Amendment Threshold by Lead Lender (Top 10)

Lead Lender	# Loans	Percent Lender Required	Supermajority Percent
Large Banks	4,234	45.55%	14.9%
Bank of America	1,165	47.20%	18.1%
JP Morgan Chase Bank	808	44.85%	11.3%
Citicorp USA	685	43.24%	10.8%
JP Morgan	507	41.89%	5.6%
Chase Manhattan Bank	379	49.35%	26.3%
Wells Fargo & Co	402	45.80%	20.6%
Credit Suisse	82	45.03%	7.2%
Barclays	54	44.15%	5.4%
Deutsche Bank	49	48.28%	6.7%
BNP Paribas	48	53.86%	45.3%
Small Banks	4,003	55.62%	39.0%
Total # Loans	8,237		

Panel A reports summary statistics of loan package and firm characteristics in our sample. All variables are defined in appendix B. Panel B reports the amendment threshold by industry. Panel C reports the amendment threshold by lead lender. *Supermajority Percent* in panel C is the percentage of loans with *Amendment Threshold (Amount)* greater than 51%.

[a] The "Supermajority Percent" in panel B and panel C is based on the full sample of 17,568 loans.

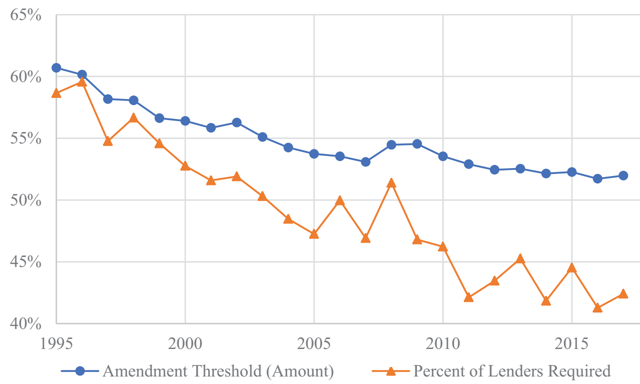

FIG. 6.—Average amendment threshold by year
This figure plots the average amendment threshold by year. Variables are defined in appendix B.

whether the group of lead lenders has than 50% of the loan commitment. When the group of lead lenders has less (more) than 50% of the loan commitment, we find that 26% (51%) of loans have a supermajority rule, and covenant waivers require agreement by at least 47% (70%) of lenders. These initial statistics are consistent with the use of voting rules to avoid undue covenant waivers by lenders who may be friendly with the borrower.

Figure 6 plots the time series patterns of the two measures of amendment thresholds. Both measures are decreasing over time. Specifically, the average of *Percent of Lenders Required* variable decreases from over 55% in the 1990s to less than 45% in the 2010s, and the average of *Amendment Threshold (Amount)* variable also decreases from 60% to around 52%.[35] Table 2 reports the summary statistics of our sample. In panels B and C of table 2, we observe significant within-industry and within-bank variation in voting thresholds, suggesting that these thresholds do not appear to reflect boilerplate practices for a given bank or industry. Further, we find that more reputable lead banks tend to have less stringent voting rules, consistent with Bushman and Wittenberg-Moerman's [2012] findings that the lead bank's reputation acts as a certification for the syndication quality, which potentially reduces the probability of agency conflicts within the syndicate.

5.3 EMPIRICAL RESULTS ON LENDER PRIVATE BENEFITS

Table 3 tests our predictions that lending agreements use more stringent voting rules when lenders have prior underwriting relationships with the borrower, and more lenient voting rules when there is a hedge fund

[35] We control for time fixed effects in all regressions. As a result, our results do not reflect any secular trends in voting thresholds

TABLE 3

Amendment Thresholds and Syndicate Relationship

Dependent Variable	Percentage of Lenders Required				Amendment Threshold (Amount)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Have underwriting relationship	0.025***				0.006***			
	(5.65)				(3.36)			
Hedge fund participant		−0.056***				−0.007*		
		(−2.97)				(−1.80)		
% Loans held by lenders with underwriting relationships			0.087***				0.025***	
			(3.74)				(2.91)	
% Loans held by hedge fund participants				−0.535***				−0.113
				(−4.48)				(−1.54)
Syndicate Relationship Intensity	−0.238***	−0.238***	−0.234***	−0.236***	−0.049***	−0.049***	−0.082***	−0.082***
	(−6.53)	(−6.54)	(−6.43)	(−6.47)	(−4.35)	(−4.35)	(−4.59)	(−4.61)
# Lenders	−0.005***	−0.005***	−0.005***	−0.005***	0.000	0.000	0.000	0.000
	(−13.75)	(−13.38)	(−13.26)	(−13.52)	(−1.62)	(−1.28)	(−0.57)	(−0.72)
Book Leverage	−0.021**	−0.018*	−0.021**	−0.017*	−0.004	−0.003	0.001	0.002
	(−2.08)	(−1.79)	(−2.04)	(−1.68)	(−1.19)	(−0.98)	(0.28)	(0.47)
ROA	−0.136*	−0.135*	−0.137*	−0.142*	−0.057***	−0.057***	−0.068**	−0.068**
	(−1.78)	(−1.76)	(−1.79)	(−1.85)	(−2.90)	(−2.86)	(−2.18)	(−2.20)

(*Continued*)

TABLE 3—*(Continued)*

Dependent Variable	Percentage of Lenders Required				Amendment Threshold (Amount)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log (Asset)	−0.005**	−0.004*	−0.005**	−0.004*	−0.006***	−0.006***	−0.006***	−0.005***
	(−2.45)	(−1.84)	(−2.31)	(−1.87)	(−7.77)	(−7.49)	(−5.43)	(−5.24)
Loan Spread	−0.011***	−0.010***	−0.011***	−0.011***	−0.002***	−0.002***	0.001	0.001
	(−4.37)	(−4.11)	(−4.40)	(−4.22)	(−2.76)	(−2.60)	(0.46)	(0.56)
Log (Package Amount)	−0.047***	−0.047***	−0.048***	−0.047***	−0.007***	−0.007***	−0.008***	−0.007***
	(−14.49)	(−14.40)	(−14.55)	(−14.40)	(−7.54)	(−7.42)	(−5.04)	(−4.92)
Log (Package Maturity)	−0.025***	−0.025***	−0.024***	−0.025***	−0.011***	−0.011***	−0.008***	−0.008***
	(−6.84)	(−6.84)	(−6.73)	(−6.86)	(−9.89)	(−9.86)	(−5.00)	(−5.10)
Observations	7,900	7,900	7,900	7,900	16,867	16,867	7,900[a]	7,900
R-squared	0.490	0.489	0.490	0.489	0.346	0.346	0.390	0.390
Year, Industry, and Lender FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

This table reports the regression results of amendment threshold on information asymmetry among lenders. Each observation is a loan package. In columns 1 to 4, the dependent variable *Percentage of Lenders Required* is the minimal number of lenders needed to reach the amendment threshold, divided by the total number of lenders. In columns 5 to 8, the dependent variable *Amendment Threshold (Amount)* is the amendment threshold stated in the contract, which is the fraction of loan principal required to amend the contract. The independent variables are described in appendix B. Robust standard errors are clustered at the borrower level in all regression specifications. Each coefficient's *t*-statistic appears directly below the coefficient estimate. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Fixed effects reduce the sample size from the 8,237 and 17,568 observations in table 2.

[a] Columns 7 and 8 have fewer observations because we require loans with non-missing lender holding data in these two columns.

in the lending syndicate (Prediction 1).[36] In column 1, the coefficient on *Have Underwriting Relationship* is 0.025 (p = 1%), suggesting that voting rules are about 0.14 standard deviations higher when lenders have a potential conflict of interest, as evidenced by prior underwriting relationships with the borrower. In column 2, the coefficient on *Hedge Fund Participant* is −0.056 (p = 1%), which is equivalent to a 0.31 standard deviation decrease in amendment thresholds. In untabulated results, we also find that, when hedge funds (hard-bargainers) have at least 20% of the loan commitment, none of these loans use a supermajority voting rule. This is consistent with voting rules being set to prevent giving the marginal vote to hedge funds, who may be hard bargainers in renegotiations.

In columns 3 and 4, we use the fraction of shares held by lenders with underwriting relationships and hedge funds as the independent variable.[37] Results are largely consistent with columns 1 and 2. With respect to economic significance, a one-standard-deviation increase in *% Loans held by lenders with underwriting relationships* is associated with a 0.041 standard deviation increase in the voting threshold, while a one-standard-deviation increase in *% Loans held by hedge fund participants* is associated with a 0.028 standard deviation decrease.[38] In columns 5 to 8, we repeat these exercises using *Amendment Threshold (Amount)* as the dependent variable and find qualitatively similar results.[39] Overall, these results support our model's predictions that voting rules are impacted by agency concerns within the lending syndicate, with tighter (more lenient) voting rules being associated with proxies of high (low) within-syndicate agency costs.

With respect to controls, we find that coefficients on *Syndicate Relationship Intensity* are negative and statistically significant across all specifications. This is consistent with both Proposition 2 and the Remark from the model—when lenders know each other well through prior syndicate relationships, voting rules are more lenient. Syndicate size (*# Lenders*) also has negative coefficients, suggesting larger syndicates tend to have more lenient voting rules. Within the context of the model, this result is consistent

[36] In untabulated results, we examine the relationship between voting rules and prior lending relationships (computed in the same manner as the prior underwriting relationships), and do not find a significant relationship between the two variables. One potential explanation is that lending relationships may capture repeated contracting, which increases contracting efficiency. The positive effect of efficiency gains from repeated contracting may offset the negative effect of higher agency costs. As a result, the net effect of the lending relationship on voting rules is unclear.

[37] Conditional on having a lender with underwriting relationship (a hedge fund participant), the fraction of the loan held by such a lender is 14.3% (8.3%).

[38] The standard deviations of *% Loans held by lenders with underwriting relationships* and *% Loans held by hedge fund participants* are 0.084 and 0.0095, respectively. Therefore, the economic magnitudes, expressed in standard deviation units, are $0.087 \times 0.084/0.179 = 0.041$ and $0.0095 \times 0.535/0.179 = 0.028$.

[39] Columns 7 and 8 have fewer observations because we require loans with nonmissing lender holding data in these two columns.

with higher renegotiation costs leading to more lenient thresholds and/or larger syndicates' additional members having Low private benefits.

5.4 EMPIRICAL RESULTS ON CONSERVATISM AND FINANCIAL COVENANTS

This subsection presents tests of our second and third predictions that voting rules are negatively associated with accounting conservatism and the number of performance covenants. Following prior literature, we use Basu's [1997] measure of asymmetric recognition of good news relative to bad news as a proxy for accounting conservatism (e.g., Beatty et al. [2008], Nikolaev [2010]). Specifically, we proxy for conservative accounting using an indicator variable that equals one when the Basu [1997] asymmetric timeliness coefficient is positive, and zero otherwise (Ball et al. [2013]).[40] Table 4, panel A, reports the results. The test neither supports nor contradicts our prediction because coefficients on *Conservatism* have the predicted negative sign but are statistically insignificant in both columns 1 and 2. This could be due to, among other things, conservative accounting influencing behaviors that are not captured in our model (Watts [2003]). For example, Bertomeu et al. [2017] show that conservatism increases manipulation, which can potentially result in a more stringent voting rule.[41]

Next, we examine the association between the financial covenants and voting rules. We use the number of performance covenants as the proxy for the number of signals in the debt contract. Performance covenants serve as tripwires on the borrower (Christensen and Nikolaev [2012]), as the income statement measures used are more informative about the current state of the firm (Saavedra [2018]).[42] Table 4, panel B, reports the results. Consistent with our model's prediction, the number of performance covenants has a negative relation with voting thresholds. In columns 2 and 4, we control for the number of capital covenants. The coefficients on performance covenants remain negative and statistically significant. Interestingly, capital covenants have a positive association with voting thresholds, suggesting that voting rules are more stringent when the loan contains more capital covenants. This is consistent with Christensen and Nikolaev's [2012] results that capital covenants, which tend to require minimum capital levels, primarily serve to align shareholders' interests with those of lenders when there is an agency problem. Likewise, our model predicts that strict voting rules mitigate agency problems between shareholders and the lending syndicate.

[40] We obtain similar results using the Basu coefficient as the independent variable.

[41] Admittedly, one limitation of Basu's conservatism measure is its lack of time-series variation, which may reduce the statistical power of the test (Bertomeu et al. [2025]).

[42] Balance sheet variables are the result of a variety of decisions that are not necessarily informative about the borrower's current performance. For example, net worth is a summary measure that includes current income, retained earnings (including big bath charges, acquisition accounting, cookie jar reserves), and dividend and payout decisions. In other words, the current performance of the firm (i.e., current net income) is only one of many components of net assets or net worth.

TABLE 4

Conservatism, Financial Covenants, and Voting Rules

Panel A.

Dependent Variables	Percentage of Lenders Required	Amendment Threshold (Amount)
	(1)	(2)
Conservatism (Basu's coefficient > 0)	−0.004	−0.001
	(−0.92)	(−0.86)
Syndicate Relationship Intensity	-0.235^{***}	-0.051^{***}
	(−5.35)	(−3.46)
Control Variables	Yes	Yes
Observations	4,906	10,076
R-squared	0.492	0.358
Year, Industry, and Lender FE	Yes	Yes

Panel B.

Dependent Variables	Percentage of Lenders Required		Amendment Threshold (Amount)	
	(1)	(2)	(3)	(4)
# Performance Covenants	-0.009^{***}	-0.006^{**}	-0.004^{***}	-0.001^{*}
	(−3.73)	(−2.39)	(−4.73)	(−1.73)
# Capital Covenants		0.010^{***}		0.011^{***}
		(3.54)		(8.66)
Syndicate Relationship Intensity	-0.230^{***}	-0.227^{***}	-0.049^{***}	-0.051^{***}
	(−5.61)	(−5.55)	(−4.02)	(−4.16)
Control Variables	Yes	Yes	Yes	Yes
Observations	7,261	7,261	15,436	15,436
R-squared	0.494	0.496	0.352	0.359
Year, Industry, and Lender FE	Yes	Yes	Yes	Yes

This table reports the regression results of amendment threshold on accounting conservatism and financial covenants. Each observation is a loan package. In panel A, the independent variable *Conservatism (Basu's coefficient* > 0) is an indicator variable equal to one if the Basu's coefficient is greater than 0 (Basu [1997]). In panel B, # *Performance Covenants* and # *Capital Covenants* are the number of performance covenants and the number of capital covenants, defined as in Christensen and Nikolaev [2012] in appendix B. Robust standard errors are clustered at the borrower level in all regression specifications. Each coefficient's *t*-statistic appears directly below the coefficient estimate. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

6. Conclusion

This study develops a model of the optimal voting rule for waiving covenants for a loan financed by lenders with heterogeneous preferences. We show that the optimal voting rule balances two costs associated with lender behavior after a covenant violation that grants lenders veto rights over a borrower's investments. First, when a false alarm trips a covenant despite the investment being profitable, lenders can extract surplus from the borrower by, for example, increasing the loan spread. Overly strict voting rules tend to give the marginal vote to lenders who will extract surplus. Second, when a covenant is tripped and the investment is indeed unprofitable, lenient lenders may waive covenants to curry favor with the managers to, for example, increase opportunities for future business. Overly lenient voting rules tend to give the marginal vote to lenders who impose no discipline on managers' postborrowing incentives to take risky but unprofitable projects. The optimal voting rule enhances contract efficiency by balancing these two costs. Our empirical tests provide evidence largely consistent with our model's predictions.

To the best of our knowledge, this paper is the first study to systematically analyze the economic forces that determine voting rules in loan contracts. The allocation of voting power has been widely studied in other settings, such as in shareholder voting and corporate governance (Harris and Raviv [1988], Sjostrom and Kim [2007], Ertimur et al. [2015]). Voting rules are important in debt contracting as they affect renegotiation considerations, which are embedded in the ex ante contract design. By developing and analyzing an extension of Gârleanu and Zwiebel [2009], this paper offers a rationale for the use of required lenders clause in loan contracts and contributes to the literature on incomplete contracting.

APPENDIX A

1.1 Derivations for Model (Sections 3 and 4)

Parametric example for private benefits: Our examples assume that, given ω, fraction α of lenders have zero private benefits and the scaled private benefits $\tilde{\delta}$ of the remaining $1-\alpha$ are lognormally distributed with $\log\tilde{\delta} \sim \mathcal{N}(\omega, \sigma^2)$, and that $\omega \sim \mathcal{N}(\bar{\omega}, \sigma_\omega^2)$. These assumptions imply that:

$$\mathrm{P}\left(F_\omega(\hat{\delta}) < 1-\hat{\mu}\right) = \mathrm{P}\left(\alpha + (1-\alpha)\,\Phi\left(\frac{\log\hat{\delta}-\omega}{\sigma}\right) < 1-\hat{\mu}\right)$$

$$= \Phi\left(-\frac{\log\hat{\delta} - \Phi^{-1}\left(\frac{1-\hat{\mu}-\alpha}{1-\alpha}\right) - \bar{\omega}}{\sigma_\omega}\right), \tag{A1}$$

where $\Phi(\cdot)$ denotes the standard normal distribution. For $\hat{\delta} > 0$, the density corresponding to $F_\omega(\hat{\delta})$ is $f_\omega(\hat{\delta}) = \frac{1-\alpha}{\hat{\delta}\sigma}\phi(\frac{\log\hat{\delta}-\omega}{\sigma})$, where $\phi(\cdot)$ denotes the density of the standard normal distribution and which satisfies the MLR condition of $\frac{\partial^2 \log f_\omega(\tilde{\delta})}{\partial\omega\partial\tilde{\delta}} = \frac{1-\alpha}{\tilde{\delta}\sigma^2} > 0$. For the plots of the density of the marginal voter with threshold $\hat{\mu}$, inverting $F_\omega(\hat{\delta}) = 1-\hat{\mu}$ gives $\log\hat{\delta} = \omega + \Phi^{-1}(\frac{1-\hat{\mu}-\alpha}{1-\alpha})\sigma$. Because $\omega \sim \mathcal{N}(\bar{\omega}, \sigma_\omega^2)$, this implies that the density of $\hat{\delta}|F_\omega(\hat{\delta}) = 1-\hat{\mu}$ is $\frac{1-\alpha}{\hat{\delta}\sigma_\omega}\phi(\frac{\log\hat{\delta}-\Phi^{-1}((1-\hat{\mu}-\alpha)/(1-\alpha))\sigma-\bar{\omega}}{\sigma_\omega})$.

Expression (A1) gives:

$$\pi_\ell(\hat{\mu}) = \underbrace{\Phi\left(\frac{\log\hat{\delta}_\ell - \Phi^{-1}\left(\frac{1-\hat{\mu}-\alpha}{1-\alpha}\right)\sigma - \bar{\omega}}{\sigma_\omega}\right)}_{\mathrm{P}\left(F_\omega(\hat{\delta}_\ell)\geq 1-\hat{\mu}\right)},$$

$$\pi_h(\hat{\mu}) = \underbrace{\Phi\left(-\frac{\log\hat{\delta}_h - \Phi^{-1}\left(\frac{1-\hat{\mu}-\alpha}{1-\alpha}\right)\sigma - \bar{\omega}}{\sigma_\omega}\right)}_{\mathrm{P}\left(F_\omega(\hat{\delta}_h)\leq 1-\hat{\mu}\right)}. \tag{A2}$$

Expression (A2) then gives $-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})} = \exp\{\frac{1}{\sigma_\omega^2}\log\frac{\hat{\delta}_h}{\hat{\delta}_\ell} \times (\Phi^{-1}(\frac{1-\hat{\mu}-\alpha}{1-\alpha})\sigma + \bar{\omega} - \log\sqrt{\hat{\delta}_h\hat{\delta}_\ell})\}$ so that:

$$-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})} = \tau \Rightarrow \hat{\mu} = (1-\alpha)\Phi\left(-\frac{1}{\sigma}\left(\log\sqrt{\hat{\delta}_h\hat{\delta}_\ell} + \frac{\log\tau}{\log(\hat{\delta}_h/\hat{\delta}_\ell)}\sigma_\omega^2 - \bar{\omega}\right)\right)$$

$$\Rightarrow \pi_\ell(\hat{\mu}) = \Phi\left(\frac{\log\sqrt{\hat{\delta}_\ell/\hat{\delta}_h} - \frac{\log\tau}{\log(\hat{\delta}_h/\hat{\delta}_\ell)}\sigma_\omega^2}{\sigma_\omega}\right),$$

$$\pi_h(\hat{\mu}) = \Phi\left(-\frac{\log\sqrt{\hat{\delta}_h/\hat{\delta}_\ell} - \frac{\log\tau}{\log(\hat{\delta}_h/\hat{\delta}_\ell)}\sigma_\omega^2}{\sigma_\omega}\right). \tag{A3}$$

Proof of Proposition 1: From (12), the first-order condition for the minimization problem is:

$$0 = \mathrm{P}(G|\underline{s})\pi'_\ell(\hat{\mu})c_r - \mathrm{P}(B|\underline{s})\pi'_h(\hat{\mu})y \Rightarrow -\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})} = \frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}. \tag{A4}$$

The ratio $-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})} = \frac{g_h(1-\hat{\mu})}{g_\ell(1-\hat{\mu})}$ where $g_i(\cdot)$ denotes the density of $F_\omega(\hat{\delta}_i)$. The assumption that $F_\omega(\hat{\delta}_h)$ dominates $F_\omega(\hat{\delta}_\ell)$ in terms of the MLR implies that $\frac{\partial}{\partial z}(\frac{g_h(z)}{g_\ell(z)}) > 0$ and $\frac{\partial}{\partial\mu}(-\frac{\pi'_h(\mu)}{\pi'_\ell(\mu)}) < 0$. This ensures a unique optimum that also satisfies the second-order condition. If $-\frac{\pi'_h(1/2)}{\pi'_\ell(1/2)} \leq \frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}$, then $-\frac{\pi'_h(\mu)}{\pi'_\ell(\mu)} > \frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}$ for all $\mu > \frac{1}{2}$ and the constraint binds. ■

Proof of Corollary 1.1: With full shareholder control, the expected cost of inefficiencies is $\mathrm{P}(B)c_r$, versus the total cost $(\mathrm{P}(G|\underline{s})\pi_\ell c_r + \mathrm{P}(B|\underline{s})\pi_h y)\mathrm{P}(\underline{s}) + \mathrm{P}(\bar{s}, B)c_r$ with a covenant. The costs with a covenant are lower if:

$$\frac{c_r}{y} > \underbrace{\pi_h(\hat{\mu}_*)}_{F_h(1-\hat{\mu}_*)} + \underbrace{\frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}}_{=-\pi'_h/\pi'_\ell} \underbrace{\pi_\ell(\hat{\mu}_*)}_{1-F_\ell(1-\hat{\mu}_*)} = F_h(1-\hat{\mu}_*) - \frac{f_h(1-\hat{\mu}_*)}{f_\ell(1-\hat{\mu}_*)}F_\ell(1-\hat{\mu}_*)$$
$$+\frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}. \tag{A5}$$

The right-hand side of (A5) is less than or equal to $\frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}$ since:

$$F_h(1-\hat{\mu}_*) = \int_0^{1-\hat{\mu}_*} \frac{f_h(\mu)}{f_\ell(\mu)} f_\ell(\mu)\mathrm{d}\mu \leq \frac{f_h(1-\hat{\mu}_*)}{f_\ell(1-\hat{\mu}_*)} \int_0^{1-\hat{\mu}_*} f_\ell(\mu)\mathrm{d}\mu$$
$$= \frac{f_h(1-\hat{\mu}_*)}{f_\ell(1-\hat{\mu}_*)} F_\ell(1-\hat{\mu}_*). \tag{A6}$$

Therefore, $\frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})} < 1$ is a sufficient, but not necessary, condition for a contract with a covenant to dominate full shareholder control. ■

Proof of Proposition 2: To show the first part, first define $\omega_i(p)$ as the ω that solves $F_\omega(\hat{\delta}_i) = p$. Then $F_\omega(\hat{\delta}_i) \leq p$ if and only if $\omega \geq \omega_i(p)$, and $\hat{\delta}_h > \hat{\delta}_\ell$ implies that $p = F_{\omega_h(p)}(\hat{\delta}_h) > F_{\omega_h(p)}(\hat{\delta}_\ell)$ so that $\omega_h(p) > \omega_\ell(p)$ since $F_\omega(\tilde{\delta})$ decreases with ω. This also implies that $\omega'_i(p) = \frac{1}{\partial F_\omega(\hat{\delta}_i)/\partial\omega} < 0$ where the derivative is evaluated at $\omega = \omega_i(p)$. The change-of-variables formula then gives:

$$-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})} = \frac{f(\omega_h(1-\hat{\mu}))}{f(\omega_\ell(1-\hat{\mu}))}\frac{\omega'_h(1-\hat{\mu})}{\omega'_\ell(1-\hat{\mu})}. \tag{A7}$$

For a parameter $\bar{\omega}$ that shifts the distribution $f(\omega)$ in terms of the MLR, we then have:

$$\frac{\mathrm{d}}{\mathrm{d}\bar{\omega}}\left(-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})}\right) = \frac{\omega'_h(1-\hat{\mu})}{\omega'_\ell(1-\hat{\mu})}\frac{\mathrm{d}}{\mathrm{d}\bar{\omega}}\left(\frac{f(\omega_h(1-\hat{\mu}))}{f(\omega_\ell(1-\hat{\mu}))}\right) > 0, \tag{A8}$$

where $\bar{\omega}$ does not affect the ω'_i terms that depend only on the conditional distributions $F_\omega(\tilde{\delta})$ and the inequality follows from the MLR property and that $\omega_h > \omega_\ell$. The implicit function theorem gives $\frac{\mathrm{d}\hat{\mu}}{\mathrm{d}\bar{\omega}} > 0$ since $-\frac{\pi'_h(\hat{\mu})}{\pi'_\ell(\hat{\mu})}$ decreases with $\hat{\mu}$. ■

Proof of remark: First, denote the notation $\tau_* = \frac{\mathrm{P}(G|\underline{s})}{\mathrm{P}(B|\underline{s})}\frac{c_r}{y}$, which is less than one by assumption, the optimized voting threshold $\hat{\mu}_*$ obtained from expression (A3) with $\tau = \tau_*, \pi^*_\ell = \pi_\ell(\hat{\mu}_*), \pi^*_h = \pi_h(\hat{\mu}_*), \bar{\sigma}^2_\omega = \frac{1}{2}\frac{\log(\hat{\delta}_h/\hat{\delta}_\ell)^2}{\log(1/\tau_*)}$, and the following where ω^*_i solves $F_{\omega^*_i}(\hat{\delta}_i) = 1-\hat{\mu}_*$:

$$\omega^*_\ell = \underbrace{\bar{\omega} + \frac{\log(1/\tau_*)}{\log(\hat{\delta}_h/\hat{\delta}_\ell)}\sigma^2_\omega - \log\sqrt{\hat{\delta}_h/\hat{\delta}_\ell}}_{>\bar{\omega}\Leftrightarrow\sigma^2_\omega>\bar{\sigma}^2_\omega}, \omega^*_h$$
$$= \underbrace{\bar{\omega} + \frac{\log(1/\tau_*)}{\log(\hat{\delta}_h/\hat{\delta}_\ell)}\sigma^2_\omega + \log\sqrt{\hat{\delta}_h/\hat{\delta}_\ell}}_{>\bar{\omega}}. \tag{A9}$$

Because $\tau_* < 1 < \frac{\hat{\delta}_h}{\hat{\delta}_\ell}$, both ω_ℓ^* and ω_h^* are increasing in σ_ω^2 with $\frac{\mathrm{d}\pi_\ell^*}{\mathrm{d}\sigma_\omega} = \frac{\mathrm{d}\pi_h^*}{\mathrm{d}\sigma_\omega} = 2\frac{\log(1/\tau_*)}{\log(\hat{\delta}_h/\hat{\delta}_\ell)}$. We can write $\hat{\mu}_*$ as $\hat{\mu}_* = (1-\alpha)\Phi(-\frac{1}{\sigma}(\omega_h^* + \log\hat{\delta}_\ell - 2\bar{\omega}))$, giving part (a). We can write $\pi_\ell^* = \Phi(\frac{\omega_\ell^*-\bar{\omega}}{\sigma_\omega})$ and $\pi_h^* = \Phi(-\frac{\omega_h^*-\bar{\omega}}{\sigma_\omega})$, giving part (b):

$$\frac{\mathrm{d}\pi_\ell^*}{\mathrm{d}\sigma_\omega} = \underbrace{-\phi(\frac{\omega_\ell^*-\bar{\omega}}{\sigma_\omega})\frac{\omega_\ell^*-\bar{\omega}}{\sigma_\omega^2}}_{\frac{\partial\pi_\ell^*}{\partial\sigma_\omega}>0\Leftrightarrow\sigma_\omega^2<\bar{\sigma}_\omega^2} + \underbrace{\phi(\frac{\omega_\ell^*-\bar{\omega}}{\sigma_\omega})\frac{1}{\sigma_\omega}}_{\frac{\partial\pi_\ell^*}{\partial\omega_\ell^*}>0}\underbrace{\frac{\mathrm{d}\omega_\ell^*}{\mathrm{d}\sigma_\omega}}_{>0}$$

$$= \phi(\frac{\omega_\ell^*-\bar{\omega}}{\sigma_\omega})\underbrace{(\frac{\mathrm{d}\omega_\ell^*}{\mathrm{d}\sigma_\omega} - \frac{\omega_\ell^*-\bar{\omega}}{\sigma_\omega})}_{>0}\frac{1}{\sigma_\omega} > 0,$$

$$\frac{\mathrm{d}\pi_h^*}{\mathrm{d}\sigma_\omega} = \underbrace{\phi(-\frac{\omega_h^*-\bar{\omega}}{\sigma_\omega})\frac{\omega_h^*-\bar{\omega}}{\sigma_\omega^2}}_{\frac{\partial\pi_h^*}{\partial\sigma_\omega}>0} \underbrace{-\phi(\frac{\omega_h^*-\bar{\omega}}{\sigma_\omega})\frac{1}{\sigma_\omega}}_{\frac{\partial\pi_h^*}{\partial\omega_h^*}<0}\underbrace{\frac{\mathrm{d}\omega_h^*}{\mathrm{d}\sigma_\omega}}_{>0}$$

$$= \phi(\frac{\omega_h^*-\bar{\omega}}{\sigma_\omega})\underbrace{(\frac{\omega_h^*-\bar{\omega}}{\sigma_\omega} - \frac{\mathrm{d}\omega_h^*}{\mathrm{d}\sigma_\omega})}_{>0\Leftrightarrow\sigma_\omega^2<\bar{\sigma}_\omega^2}\frac{1}{\sigma_\omega}. \tag{A10}$$

For part (c), direct computations give $\frac{\mathrm{d}\pi_h^*/\mathrm{d}\sigma_\omega}{\mathrm{d}\pi_\ell^*/\mathrm{d}\sigma_\omega} = \tau * \frac{\bar{\sigma}_\omega^2-\sigma_\omega^2}{\bar{\sigma}_\omega^2+\sigma_\omega^2}$ so that:

$$\frac{\mathrm{dPayoff}}{\mathrm{d}\sigma_\omega} = -\left(\mathrm{P}(G|\underline{s})c_r\frac{\mathrm{d}\pi_\ell^*}{\mathrm{d}\sigma_\omega} + P(B|\underline{s})y\frac{\mathrm{d}\pi_h^*}{\mathrm{d}\sigma_\omega}\right)\mathrm{P}(\underline{s})$$

$$= -\underbrace{\left(\tau_* + \frac{\mathrm{d}\pi_h^*/\mathrm{d}\sigma_\omega}{\mathrm{d}\pi_\ell^*/\mathrm{d}\sigma_\omega}\right)}_{>0}\frac{\mathrm{d}\pi_\ell^*}{\mathrm{d}\sigma_\omega}\mathrm{P}(B,\underline{s})y < 0, \tag{A11}$$

where the inequality follows because $\bar{\sigma}_\omega^2 > 0$ implies that $\frac{\mathrm{d}\pi_h^*/\mathrm{d}\sigma_\omega}{\mathrm{d}\pi_\ell^*/\mathrm{d}\sigma_\omega} > -\tau_*$. ■

Proof of Proposition 3: If the $\hat{\mu} \geq \frac{1}{2}$ constraint binds, then the voting rule does not vary with conservatism. Otherwise, applying the implicit function theorem to the first-order condition to (A4) and using $\frac{\partial}{\partial\mu}(-\frac{\pi_h'(\mu)}{\pi_\ell'(\mu)}) < 0$ implies that the sign of $\frac{\mathrm{d}\hat{\mu}}{\mathrm{d}c}$ is the opposite the sign of:

$$\frac{\partial}{\partial c}\left(\frac{\mathrm{P}(\underline{s}|G)}{\mathrm{P}(\underline{s}|B)}\frac{\mathrm{P}(G)}{\mathrm{P}(B)}\frac{c_r}{y}\right) = \frac{\mathrm{P}(G)}{\mathrm{P}(B)}\frac{c_r}{y}\frac{\partial}{\partial c}\left(\frac{\mathrm{P}(\underline{s}|G)}{\mathrm{P}(\underline{s}|B)}\right) > 0, \tag{A12}$$

where the inequality follows from the assumption that $\frac{\mathrm{P}(s|G)}{\mathrm{P}(s|B)}$ increases in c. ■

Proof of Proposition 4: From expression (13) and the assumption that $-\frac{\pi_h'(\mu)}{\pi_\ell'(\mu)}$ decreases in μ, the threshold $\hat{\mu}_M$ with M covenants is less than the threshold $\hat{\mu}_{M-1}$ with $M-1$ covenants if $\frac{1-\mathrm{P}(\bar{S}_M|G)}{1-\mathrm{P}(\bar{S}_M|B)} > \frac{1-\mathrm{P}(\bar{S}_{M-1}|G)}{1-\mathrm{P}(\bar{S}_{M-1}|B)}$. Before proceeding, we define the ratio $\kappa_M = \frac{\mathrm{P}(\bar{S}_M|B)}{\mathrm{P}(\bar{S}_M|G)} \in (0,1)$ The assumption

$\mathrm{P}(\underline{s}_M|\bar{S}_{M-1},B) > \mathrm{P}(\underline{s}_M|\bar{S}_{M-1},G)$ is equivalent to $\frac{\mathrm{P}(\bar{s}_M|\bar{S}_{M-1},B)}{\mathrm{P}(\bar{s}_M|\bar{S}_{M-1},G)} < 1$. This then implies that $\kappa_M = \frac{\mathrm{P}(\bar{S}_M|\bar{S}_{M-1},B)}{\mathrm{P}(\bar{S}_M|\bar{S}_{M-1},G)}\frac{\mathrm{P}(\bar{S}_{M-1}|B)}{\mathrm{P}(\bar{S}_{M-1}|G)}$ is decreasing with m. The condition $\frac{1-\mathrm{P}(\bar{S}_M|G)}{1-\mathrm{P}(\bar{S}_M|B)} > \frac{1-\mathrm{P}(\bar{S}_{M-1}|G)}{1-\mathrm{P}(\bar{S}_{M-1}|B)}$ is equivalent to:

$$\underbrace{1-\mathrm{P}(\bar{S}_M|\bar{S}_{M-1},G)}_{\mathrm{P}('\mathrm{Falsealarm}')} > \frac{1}{1+\frac{1-\kappa_{M-1}}{(\kappa_{M-1}-\kappa_M)(1-\mathrm{P}(S_{M-1}|G))}}, \tag{A13}$$

giving the first part of the proposition. For the second part, with IID signals $1-\mathrm{P}(\bar{S}_M|\bar{S}_{M-1},G) = 1-\mathrm{P}(\bar{s}|G)$, $\kappa_M = (\frac{\mathrm{P}(\bar{s}|G)}{\mathrm{P}(\bar{s}|B)})^M = \kappa^M$, and:

$$\underbrace{\frac{1-\kappa^{M-1}}{(\kappa^{M-1}-\kappa^M)\left(1-\mathrm{P}(\bar{s}|G)^{M-1}\right)}}_{\frac{1-\kappa_{M-1}}{(\kappa_{M-1}-\kappa_M)(1-\mathrm{P}(S_{M-1}|G))}} \geq \frac{M-1}{1-\mathrm{P}(\bar{s}|G)^{M-1}} \geq \frac{1}{1-\mathrm{P}(\bar{s}|G)}, \tag{A14}$$

where Bernoulli's inequality implies left-hand side of (A14) decreases with κ and therefore exceeds its limit $\frac{M-1}{1-\mathrm{P}(\bar{s}|G)^{M-1}}$ as $\kappa \to 1$.[43] Bernoulli's inequality also gives $\frac{M-1}{1-\mathrm{P}(\bar{s}|G)^{M-1}} \geq \frac{1}{1-\mathrm{P}(\bar{s}|G)}$. Expression (A14) implies that the right-hand side of (A13) is less than $\frac{1-\mathrm{P}(\bar{s}|G)}{2-\mathrm{P}(\bar{s}|G)}$, which is less than $1-\mathrm{P}(\bar{s}|G)$ so that (A13) always holds with IID signals. ■

APPENDIX B

1.1 Variable Descriptions

Variable	Description
Amendment Threshold (Amount)	Minimal percent of loan amount required for loan amendment, specified in the loan contract
Percent of Lenders Required	Minimal # lenders required to reach the amendment threshold divided by total # lenders
Have Underwriting Relationship	An indicator variable equal to one if one of the lenders was also a lead underwriter for the borrower in prior security issuances in the past three years. Dealscan sometimes have slightly different names for the same bank. We aggregate these banks in appendix C
Hedge fund participant	An indicator variable equal to one if the non–lead lender is a hedge fund. We use Hedge Fund Research (HFR) database to identify the list of hedge fund lenders

[43] Specifically, $x^m \geq 1 + m(x-1)$ for $x \geq 0$, and the derivative of (A14) has the same sign as $1 + m(\kappa - 1) - \kappa^m$.

Variable	Description
Syndicate Relationship Intensity	The average number of deals arranged by the lead bank with at least one of the current participants, measured over a three-year horizon and expressed as a percent of the total deals underwritten during this period. Dealscan sometimes have slightly different names for the same bank. We aggregate these banks in appendix C
# Lenders	Number of lenders in the syndicate
Book Leverage	Total debt/total assets, measured at the quarter end before the loan starting date
ROA	Net income/total assets, measured at the quarter end before the loan starting date
Log (Asset)	Log Asset, measured at the quarter end before the loan starting date
Log (Amount)	Log Loan Amount. If a loan package has multiple facilities, the amount is the sum of all facility amounts
Log (Maturity)	Log Loan Maturity. If a loan package has multiple facilities, the package maturity is maximum maturity of all facilities
Loan Spread	The spread the borrower pays in basis points over LIBOR for each dollar drawn down. If a loan package has multiple facilities, the spread is calculated as the average spread of all facilities
Conservatism	An indicator variable equal to one if the Basu's coefficient is greater than zero. We estimate the Basu's coefficient using the following firm-level regression: $E_t/P_{t-1} = b_0 + b_1 * D(Ret_t < 0) + b_2 * Ret_t + b_3 * Ret_t * D(Ret_t < 0)$, where E_t/P_{t-1} is the earnings per share excluding extraordinary items of the year divided by the stock price at the previous year end. $D(Ret_t < 0)$ is an indicator variable if annual return is less than 0. Ret_t is the market-adjusted annual return of the firm. For each loan, we estimate the Basu's coefficient of the borrower using the data for the past 10 years, and require each regression to have at least three years of nonmissing data
# Performance Covenants	Number of performance covenants, where performance covenants include interest coverage, cash interest coverage, fixed charge coverage, debt to EBITDA, Debt service coverage and minimum EDBITDA covenants
# Capital Covenants	Number of capital covenants, where capital covenants include net worth, leverage ratio, debt to equity, debt to tangible net worth, current ratio, and quick ratio covenants
Reputable Bank	Bank has more than 2% market share in the syndicated loan market in the sample period (Bushman and Wittenberg-Moerman [2012])

APPENDIX C

1.1 Lender Id Aggregation

This table aggregates lenders with slightly different names that are likely to belong to the same parent bank. While this list is by no means capturing all variations in lenders' names in Dealscan, it covers a majority of loans in our sample.

Old Lender ID	Old Lender Name	New Lender ID	New Lender Name
127349	Bank of America Merrill Lynch	6532	Bank of America
84685	Bank of America NA	6532	Bank of America
68290	Citibank NA	5893	Citibank
6231	Citicorp North America Inc	5893	Citibank
10432	Citicorp USA Inc	5893	Citibank
30949	Citigroup	5893	Citibank
113896	Credit Suisse Cayman Islands	7828	Credit Suisse AG
1468	Credit Suisse First Boston	7829	Credit Suisse AG
99326	Deutsche Bank AG New York Branch	7861	Deutsche Bank AG
14037	Fleet Bank NA	6225	Fleet Bank
5972	Fleet National Bank	6225	Fleet Bank
6531	JP Morgan & Co	38939	JP Morgan
87488	JP Morgan Chase	87000	JP Morgan Chase Bank NA
83272	JP Morgan Chase & Co	87000	JP Morgan Chase Bank NA
84950	JPMorgan Chase Bank	87000	JP Morgan Chase Bank NA
28124	Key Bank NA	22334	Key Bank
27114	Keybank NA	22334	Key Bank
87225	PNC Bank NA	6542	PNC Bank
13252	Toronto Dominion Bank [Texas]	7827	Toronto Dominion Bank
87229	Wachovia Bank NA	6443	Wachovia Bank
6541	Wells Fargo & Co	6123	Wells Fargo Bank
87407	Wells Fargo Bank NA	6123	Wells Fargo Bank

APPENDIX D

Example of Required Lenders Clause

Below is an excerpt from a loan agreement between THQ Inc. and a group of lenders. The loan is syndicated by Bank of America.[44]

> *"Required Lenders": Lenders (subject to Section 4.2. Defaulting Lenders) having (a) Revolver Commitments in excess of 50% of the aggregate Revolver Commitments; and (b) if the Revolver Commitments have terminated, Loans in excess of 50% of all outstanding Loans.*

14.1. Consents, Amendments, and Waivers
14.1.1. Amendment. No modification of any Loan Document, including any extension or amendment of a Loan Document or any waiver of a Default or Event of Default, shall be effective without the prior written agreement of Agent (with the consent of Required Lenders) and each Obligor party to such Loan Document; provided, however, that

a. *without the prior written consent of Agent, no modification shall be effective with respect to any provision in a Loan Document that relates to any rights, duties or discretion of Agent;*

[44] https://www.sec.gov/Archives/edgar/data/865570/000110465909062958/a09-31246_1ex10d1.htm

b. *without the prior written consent of Issuing Bank, no modification shall be effective with respect to any LC Obligations or Section 2.3;*
c. *without the prior written consent of each affected Lender, no modification shall be effective that would (i) increase the Commitment of such Lender; or (ii) reduce the amount of, or waive or delay payment of, any principal, interest or fees payable to such Lender; and*
d. *without the prior written consent of all Lenders (except a Defaulting Lender as provided in section* 4.2*), no modification shall be effective that would (i) extend the Revolver Termination Date; (ii) alter Section 5.6, 7.1 (except to add Collateral) or 14.1.1; (iii) amend the definitions of Borrowing Base (and the defined terms used in such definition), Pro Rata or Required Lenders; (iv) increase any advance rate or increase total Commitments; (vi) release Collateral with a book value greater than \$10,000,000 during any calendar year, except as currently contemplated by the Loan Documents; or (vii) release any Obligor from liability for any Obligations, if such Obligor is Solvent at the time of the release.*

REFERENCES

Aghion, P., and P. Bolton. “An Incomplete Contracts Approach to Financial Contracting.” *Review of Economic Studies* 59 (1992): 473–94.

Asquith, P.; A. Beatty; and J. Weber. “Performance Pricing in Bank Debt Contracts.” *Journal of Accounting and Economics* 40 (2005): 101–28.

Armstrong, C.; W. Guay; and J. Weber. “The Role of Information and Financial Reporting in Corporate Governance and Debt Contracting.” *Journal of Accounting and Economics* 50 (2010): 179–234.

Ball, R.; S. Kothari; and V. Nikolaev. “Econometrics of the Basu Asymmetric Timeliness Coefficient and Accounting Conservatism.” *Journal of Accounting Research* 51 (2013): 1071–97.

Basu, S. “The Conservatism Principle and the Asymmetric Timeliness of Earnings.” *Journal of Accounting and Economics* 24 (1997): 3–37.

Beatty, A.; J. Weber; and J. Yu. “Conservatism and Debt.” *Journal of Accounting and Economics* 45 (2008): 154–74.

Beneish, M., and E. Press. “Costs of Technical Violation of Accounting-Based Debt Covenants.” *The Accounting Review* 68 (1993): 233–57.

Berlin, M., and L. Mester. “Debt Covenants and Renegotiation.” *Journal of Financial Intermediation* 2 (1992): 95–133.

Bertomeu, J.; E. Cheynel; E. Y. Liao; and M. Milone. “Using Machine Learning to Measure Conservatism.” *Management Science* 71 (2025): 1504–22.

Bertomeu, J.; M. Darrough; and W. Xue. “Optimal Conservatism with Earnings Manipulation” *Contemporary Accounting Research* 34 (2017): 252–84.

Bharath, S.; S. Dahiya; A. Saunders; and A. Srinivasan. “So What Do I Get? The Bank’s View of Lending Relationships.” *Journal of Financial Economics* 85 (2007): 368–419.

Bharath, S.; S. Dahiya; A. Saunders; and A. Srinivasan. “Lending Relationships and Loan Contract Terms.” *Review of Financial Studies* 24 (2011): 1141–203.

Bolton, P., and S. Scharfstein. “Optimal Debt Structure and the Number of Creditors.” *Journal of Political Economy* 104 (1996): 1–25.

Bushman, R., and R. Wittenberg-Moerman. “The Role of Bank Reputation in ‘Certifying’ Future Performance Implications of Borrowers’ Accounting Numbers.” *Journal of Accounting Research* 50 (2012): 883–930.

Bushman, R.; A. Smith; and R. Wittenberg-Moerman. "Price Discovery and Dissemination of Private Information by Loan Syndicate Participants." *Journal of Accounting Research* 48 (2010): 921–72.

Bushman, R.; S. Sohn; and J. Zhao. "Partisan Banks and Creditor Coordination Within Loan Syndicates." (2025): Available at SSRN: https://ssrn.com/abstract=4778165.

Caskey, J.; K. Huang; and D. Saavedra. "Noncompliance with SEC Regulations Evidence from Timely Loan Disclosures." *Review of Accounting Studies* 28 (2023): 126–63.

Caskey, J., and V. Laux. "Corporate Governance, Accounting Conservatism, and Manipulation." *Management Science* 63 (2017): 424–37.

Champagne, C., and L. Kryzanowski. "Are Current Syndicated Loan Alliances Related to Past Alliances?" *Journal of Banking & Finance* 31 (2007): 3145–61.

Chava, S., and M. Roberts. "How Does Financing Impact Investment? The Role of Debt Covenants." *The Journal of Finance* 63 (2008): 2085–121.

Chen, K., and K. Wei. "Creditors' Decisions to Waive Violations of Accounting-Based Debt Covenants." *The Accounting Review* 68 (1993): 219–32.

Christensen, H., and V. Nikolaev. "Capital Versus Performance Covenants in Debt Contracts." *Journal of Accounting Research* 50 (2012): 75–116.

Christensen, H.; V. Nikolaev; and R. Wittenberg-Moerman. "Accounting Information in Financial Contracting: The Incomplete Contract Theory Perspective." *Journal of Accounting Research* 54 (2016): 397–435.

Dass, N.; V. Nanda; and Q. Wang. "Within-Syndicate Conflicts, Loan Covenants, and Syndicate Formation." *Financial Management* 49 (2020): 547–83.

DeFond, M., and J. Jiambalvo. "Debt Covenant Violation and Manipulation of Accruals." *Journal of Accounting and Economics* 17 (1994): 145–76.

Demerjian, P. "Accounting Standards and debt Covenants Has the 'Balance Sheet Approach' Led to a Decline in the Use of Balance Sheet Covenants?" *Journal of Accounting and Economics* 52 (2011): 178–202.

Demerjian, P. "Uncertainty and Debt Covenants." *Review of Accounting Studies* 22 (2017): 1156–97.

Demerjian, P., and E. Owens. "Measuring the Probability of Financial Covenant Violation in Private Debt Contracts." *Journal of Accounting and Economics* 61 (2016): 433–47.

Demerjian, P., E. Owens, and M. Sokolowski. "Lender capital management and financial covenant strictness." *The Accounting Review* 98 (2023): 146–72.

Denis, D., and J. Wang. "Debt Covenant Renegotiations and Creditor Control Rights." *Journal of Financial Economics* 113 (2014): 348–67.

Diamond, D. "Monitoring and reputation: The choice between bank loans and directly placed debt." *Journal of Political Economy* 99 (1991): 689–721.

Dichev, I., and D. Skinner. "Large-Sample Evidence on the Debt Covenant Hypothesis." *Journal of Accounting Research* 40 (2002): 1091–123.

Dou, Y. "The Debt-Contracting Value of Accounting Numbers and Financial Covenant Renegotiation." *Management Science* 66 (2020): 1124–48.

Drucker, S., and M. Puri. "On the Benefits of Concurrent Lending and Underwriting." *The Journal of Finance* 60 (2005): 2763–99.

Ertimur, Y.; F. Ferri; and D. Oesch. "Does the Director Election System Matter? Evidence from Majority Voting." *Review of Accounting Studies* 20 (2015): 1–41.

Fernando, C.; A. May; and W. Megginson. "The Value of Investment Banking Relationships: Evidence from the Collapse of Lehman Brothers." *The Journal of Finance* 67 (2012): 235–70.

Gârleanu, N. and J. Zwiebel. "Design and Renegotiation of Debt Covenants." *Review of Financial Studies* 22 (2009): 749–81.

Gigler, F.; C. Kanodia; H. Sapra; and R. Venugopalan. "Accounting Conservatism and the Efficiency of Debt Contracts." *Journal of Accounting Research* 47 (2009): 767–97.

Guttman, I., and I. Marinovic. "Debt Contracts in the Presence of Performance Manipulation." *Review of Accounting Studies* 23 (2018): 1005–41.

Harris, M., and A. Raviv. "Corporate Governance Voting Rights and Majority Rules." *Journal of Financial Economics* 20 (1988): 203–35.

Hart, O. "Financial Contracting." *Journal of Economic Literature* 39 (2001): 1079–100.

Ivashina, V. "Asymmetric Information Effects on Loan Spreads." *Journal of Financial Economics* 92 (2009): 300–319.

Jeon, J.; C. Lee; T. Nasser; and M. Via. "Multiple Lead Underwriter IPOs and Firm Visibility." *Journal of Corporate Finance* 32 (2015): 128–49.

Jiang, W.; K. Li; and W. Wang. "Hedge Funds and Chapter 11." *The Journal of Finance* 67 (2012): 513–60.

Li, N.; F. Vasvari; and R. Wittenberg-Moerman. "Dynamic Threshold Values in Earnings-Based Covenants." *Journal of Accounting and Economics* 61 (2016): 605–29.

Li, X. "Relationship Lending in Syndicated Loans: A Participant's Perspective." (2021) Available at SSRN: https://ssrn.com/abstract=3716948.

Li, E.; M. Neamtiu; and Z. Tu. "Do Firms Withhold Loan Covenant Details?" *The Accounting Review* 99 (2024): 421–49.

LSTA. *The Handbook of Loan Syndications and Trading*, Second edition. New York: McGraw Hill, 2016.

Luneva, I. "Explaining Debt Covenant Amendments: A Structural Approach." 2025: Available at SSRN: https://ssrn.com/abstract=4588878.

Magee, R., and S. Sridhar. "Financial Contracts, Opportunism, and Disclosure Management." *Review of Accounting Studies* 1 (1996): 225–58

Martin, X.; X. Tang; H, Wang; and Y. Xie. "Lender Coordination and Loan Renegotiation." 2024: Available at SSRN: https://ssrn.com/abstract=4484738.

Nikolaev, V. "Debt Covenants and Accounting Conservatism." *Journal of Accounting Research* 48 (2010): 137–76.

Nikolaev, V. "Scope for Renegotiation in Private Debt Contracts." *Journal of Accounting and Economics* 65 (2018): 270–301.

Nini, G.; D. Smith; and A. Sufi. "Creditor Control Rights, Corporate Governance, and Firm Value." *The Review of Financial Studies* 25 (2012): 1713–61.

Proskauer. "Navigating the Club in Private Credit Deals." 2020: Available at https://www.proskauer.com/alert/navigating-the-club-in-private-credit-deals.

Prilmeier, R. "Why Do Loans Contain Covenants? Evidence from Lending Relationships." *Journal of Financial Economics* 123 (2017): 558–79.

Rajan, R., and A. Winton. "Covenants and Collateral as Incentives to Monitor." *The Journal of Finance* 50 (1995): 1113–46.

Roberts, M. "The Role of Dynamic Renegotiation and Asymmetric Information in Financial Contracting." *Journal of Financial Economics* 116 (2015): 61–81.

Roberts, M., and A. Sufi. "Control Rights and Capital Structure: An Empirical Investigation." *The Journal of Finance* 64 (2009a): 1657–95.

Roberts, M., and A. Sufi. "Renegotiation of Financial Contracts: Evidence from Private Credit Agreements." *Journal of Financial Economics* 93 (2009b): 159–84.

Saavedra, D. "Syndicate Size and the Choice of Covenants in Debt Contracts." *The Accounting Review* 93 (2018): 301–29.

Saavedra, D. "Do Firms Follow the SEC's Confidential Treatment Protocols? Evidence from Credit Agreements." *Review of Accounting Studies* 28 (2023): 1388–412.

Sjostrom Jr., W., and W. Kim. "Majority Voting for the Election of Directors." *Connecticut Law Review* 40 (2007): 459–510.

Smith Jr, C. "A Perspective on Accounting-Based Debt Covenant Violations." *The Accounting Review* 68 (1993): 289–303.

Sufi, A. "Information Asymmetry and Financing Arrangements: Evidence from Syndicated Loans." *The Journal of Finance* 62 (2007): 629–68.

Sweeney, A. "Debt-Covenant Violations and Managers' Accounting Responses." *Journal of Accounting and Economics* 17 (1994): 281–308.

Vashishtha, R. "The Role of Bank Monitoring in Borrowers' Discretionary Disclosure: Evidence from Covenant Violations." *Journal of Accounting and Economics* 57 (2014): 176–95.

Watts, R. "Conservatism in Accounting Part I: Explanations and Implications." *Accounting Horizons* 17 (2003): 207–21.

WHITNEY, N., and M. ZELINSKY. "Pro Rata Sharing Provisions in Credit Agreements: What Lenders and Loan Investors Need to Know." *Pratt's Journal of Bankruptcy Law* 13 (2017): 384–86.

ZHONG, H. "A Dynamic Model of Optimal Creditor Dispersion." *The Journal of Finance* 76 (2021): 267–316.

ZHU, C. "Treatment of Accounting Changes and Covenant Violation Errors." *Journal of Accounting Research* 62 (2024): 783–824.